

Observation Disciplines

How logics are generated from term structure,
and why graded where-clauses are the mortal scientist’s instrument

An omnibus

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Abstract

A logic, on the account developed here, is not a thing one chooses and then applies to a computational system. It is a thing the system *generates*. Fix a term language and its rewrite rules, fix a container in which witnesses are aggregated, and fix an algebra in which observations take their values, and the connectives follow: the term constructors lift to structural predicates, the contexts labelling rewrites lift to modalities, and the container’s own operations supply whatever propositional structure the logic is going to have. Turning those dials does not produce variations on one logic. It produces a range of genuinely different *observation disciplines* over the same term structure — disciplines that differ in what they can distinguish, what they cost to evaluate, and what they are able to learn. Classical linear logic, intuitionistic logic, the logic of actionable coalitions, spatial-behavioural logic, and the fuzzy and complex-valued logics of graded execution are exhibited here as settings of the same generator rather than as separate inventions.

The second half of the thesis concerns placement. The same formula can be used as a specification checked from outside, as a guard evaluated at the redex, or as a hypothesis revised over time. It is the second placement — the *where-clause* of a for-comprehension — that turns a logic from a description into an instrument, because it is there that the formula’s value resolves the system’s own non-determinism. Allowing that value to be graded rather than crisp is a small syntactic change with large consequences: it gives a resource-bounded learner a coordinate in which to record disappointment, and so a way to revise a belief without revising a formula.

This is a summary document. It assembles results established in a series of research notes, restates them in a common vocabulary, and argues that they are one result seen from several sides. Where something new is claimed, it is flagged as such and usually left as a conjecture.

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Reader’s guide

This document sprawls, deliberately and in a known tradition. It is meant to be entered at several points.

Part 0 states the whole argument in about ten pages with no category theory and no formal semantics. A reader who wants to know what is being claimed and why it might matter should read Part 0 and stop; everything after it is elaboration and evidence. Every subsequent part opens with a plain-language paragraph, set off before any symbols appear, saying what that part establishes.

Parts I and II are the generation story. Part I gives the generator: what a logic is made of, which parameters it has, and in what sense the output is not *ad hoc*. Part II works the parameter that is least familiar — the container in which witnesses are collected — and shows that the substructural hierarchy, intuitionistic logic, classical linear logic and the logic of decentralised consensus are four settings of it.

Parts III to V are the instrument story: where a formula is placed (Part III), what happens when its truth values are graded (Part IV), and what resolving the resulting non-determinism costs (Part V).

Parts VI to VIII are the evidence. Part VI walks six observation disciplines with measured examples drawn from games, foraging, quantum circuits, stampeding herds and honeybee swarms. Part VII is the learning argument proper and the fractal it lives in. Part VIII inventories what is shown, what is not, and what is open.

Sources. Nothing here is offered as a replacement for the notes it summarises. The technical arguments live in [1, 2, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 4, 6], and each part names which of them it is compressing.

Two notational conventions, stated once. The reflective higher-order calculus is written rho calculus, never with the Greek letter that follows π . The name is an acronym and a pun at the same time — r.h.o. for *reflective higher order*, and the transliteration of the letter after π , marking the calculus as the successor of the π -calculus — and writing it as a Greek letter loses both. Quotation is written $@P$ and dereference $*x$; the corner-quote notation of the 2005 papers is not used.

Part 0 — The argument in ten pages

This part uses no category theory and states no theorems. It is the whole argument, in the order in which it is best met for the first time.

0.1 The question

Suppose you want to build something that learns about its world. Not a function approximator fitted offline to a corpus, but a thing that is *in* a world, that spends something to look at it, that forms guesses and finds out whether they were right, and that dies if it spends more than it gets back.

Very early in the design, before any question about architecture, a question arrives that is easy to skip past: *what can this thing say?* Whatever it believes, it believes in some vocabulary. Whatever hypothesis it entertains, that hypothesis is a sentence in some language. Whatever experiment it runs, the experiment's outcome has to be expressible as a change in that language's truth values. The vocabulary is not decoration on the learner. It is the shape of everything the learner will ever be able to think.

The usual move is to pick one. Take first-order logic, or a probabilistic graphical model, or a set of hand-designed features, and then get on with the interesting work. The move is so standard that it barely registers as a move. But it has a cost, and the cost is that the vocabulary is now unaccountable: it floats above the machine, related to it by nothing in particular, and when it turns out to be the wrong vocabulary there is no principled way to say what a better one would be.

The claim of this document is that the choice can be taken out of your hands, in a good way. If you have already said what your computational system is — what its states look like and how it steps — then you have *already said* what its logic is. The connectives are recoverable from the syntax. The modalities are recoverable from the rewrite rules. There is nothing left to design. And what falls out is not one logic but a structured range of them, because the recovery procedure has a small number of parameters, and different settings of those parameters give different — genuinely, usefully different — disciplines of observation over the same underlying system.

We call these *observation disciplines*, and the plan of this document is to exhibit the range, show that some famous logics are points in it, and then argue that one particular way of deploying such a discipline turns it from a description into an instrument that a resource-bounded learner can actually hold.

0.2 What a logic is made of

Begin from a commitment that sounds technical but is really a discipline of honesty: *a formula denotes the collection of computations that witness it*. This is realizability, in the sense that goes back to Kleene [30]. It says: do not tell me a formula is true; show me what makes it true, and let it be a thing that could run.

Adopt that, and a logic falls into three parts. Not by stipulation — by inspection. Every connective you can build is of one of three kinds.

Structural connectives come from the term constructors. Your terms are built by constructors: a pairing, a send, a receive, a parallel composition. Each of them lifts to a predicate about *shape*. If terms can be put side by side — $P \mid Q$ — then there is a connective $\varphi \mid \psi$ which holds of a term exactly when the term can be split into a part satisfying φ and a part satisfying ψ . If terms can send on a channel, there is a connective saying *is a send on a channel satisfying* These are not invented; there is exactly one per constructor, and you get them all whether you want them or not.

Behavioural connectives come from the rewrite rules. Your terms step. Each step happens somewhere: there is a surrounding context, and a redex sitting in it. Label the step by that context and you have a modality: $\langle K \rangle \varphi$ means *there is a step, at the position marked by K , after which φ holds*. This is the Hennessy–Milner construction [25] with contexts in place of action labels, which turns out to matter a great deal: an action-labelled logic looking at a board game sees one label, “a move was made,” while a context-labelled logic sees as many labels as there are legal moves.

Collection connectives come from the container. A formula denotes a collection of witnesses — but a collection of *what kind*? A set? A multiset? A list? A probability distribution? This is a real choice, and it is the one that carries the propositional logic. If witnesses live in sets, then two formulae can be intersected and united, and you get conjunction and disjunction in their familiar form. If witnesses live in multisets, you can no longer silently discard a duplicate, and what you get instead is a *multiplicative* conjunction that tracks how many times a resource was used. The container’s algebra is the logic’s structural rules. This is not an analogy; it is an identification, and Part I states it precisely.

So a logic has three components, and two of them — structural and behavioural — are read off the presentation of the system with no discretion at all. The third is a genuine parameter. There is a fourth parameter, introduced below.

0.3 Turning the dials

Here is the payoff of decomposing rather than choosing. The components are dials, and turning them lands you on logics you already know, plus many you do not.

Turn the container dial from sets to multisets and the structural rules change: you lose the right to use a witness twice, and what emerges is the multiplicative fragment of *linear logic*. Stack a powerset on top of a free commutative monoid and add a distinguished element to serve as a pole, and what emerges is classical linear logic in full, with its two conjunctions and its exponential [17, 18]. Part II works this instance carefully, because it is the one that most convincingly makes the case: linear logic is famously the logic of resources, and here it is not *modelled* as a logic of resources but *generated* from a container that keeps count.

Set the container to the open sets of a topology and you get intuitionistic logic in its geometric guise, where a proposition is an affirmation, something that can be positively said [32]. Now weaken the container by dropping one axiom — let the opens be closed under arbitrary unions but not under intersection — and you get Gabbay’s *semiotopology* [33], whose points are participants and whose opens are coalitions with the resources to act on their own. The intersection of two coalitions each strong enough to act need not itself be strong enough to act. That single dropped

axiom is the whole difference from topology, and the logic it generates is the logic of decentralised consensus. Conjunction does not survive as an operation; it survives as a *relation*, “these two have a witness in common,” and Gabbay arrived at exactly that relation independently, for reasons of algebraic duality [34]. Two routes, one relation. That coincidence is the best single piece of evidence that the generation story is tracking something real rather than reorganising things we already knew how to write down.

The fourth dial is the one this document is most interested in, and it does not appear in classical treatments at all: *what does an observation return?* The default answer is a bit — true or false. But nothing forces it. Let a formula return a number in $[0, 1]$, or a non-negative real, or a value in a tropical algebra where addition is minimum, or a complex amplitude. Each choice is a coherent logic; each has different laws; and, as we will see, the choice determines what a learner using that logic is capable of.

Slogan 0.3.1. *A logic is a choice of observation discipline over a term language: what shapes you may name, what steps you may see, how witnesses are collected, and what an observation returns.*

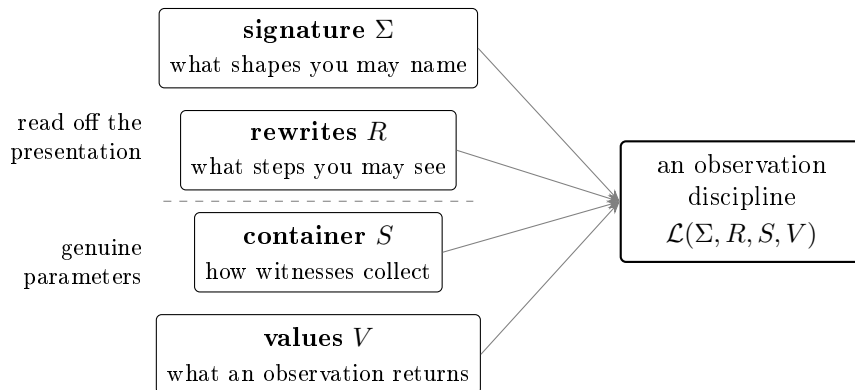


Figure 1: The four dials. The first two are read off the presentation of the system; the last two are where the design freedom lives.

0.4 Three places to put a formula

Having a logic is not yet having an instrument. The same formula can be used in three quite different ways, and only one of them makes it something a system can look *through*.

As a specification. Write $\vdash \varphi \triangleright P$: from the formula, synthesise a program that realises it. This is the proofs-as-programs direction, and it is the direction most of the literature on logics for concurrency takes. It is also the expensive direction, because it requires a proof theory — and for the logics generated here, the proof theory is usually the open problem.

As a test. Write $P \models \varphi$: given a program, check whether the formula holds. This needs only the semantics, which the generation procedure hands you for free. It is dramatically cheaper than synthesis, and it is what an observer does.

As a guard. Write the formula *inside the program*, on the binder:

`for( y <- x where  $\varphi$  )P`

Now the formula is evaluated at the moment a communication is about to happen, against the candidate message, and its value decides whether the communication occurs. This is the placement that makes a logic an instrument. The formula is no longer a description of the computation; it is part of the computation, and it is the part that decides what the computation does next.

That last point deserves emphasis, because it is the hinge of the whole document. A concurrent system of any interest is non-deterministic: at any moment several communications could occur, and something has to decide which does. In a conventional implementation that decision is made by the hardware — by which core happened to win a race. It is made, in other words, by something the language has nothing to say about. Putting a formula in the binder moves the decision *into the language*. The guard’s value is what resolves the race. And since the guard is a formula of the generated logic, the resolution of non-determinism becomes a matter of *what the system is able to observe about itself*.

0.5 Why a Boolean guard cannot learn

Now grade the guard, and watch what happens.

A crisp guard is a partition: it splits the candidate messages into those that pass and those that do not. A graded guard is a map into a value algebra: each candidate gets a number, and the numbers become propensities — how likely, or how strongly weighted, each possible communication is.

The difference sounds like a convenience. It is not. Consider a small foraging organism from [2]. It lives in a world containing specimens of two kinds: rich ones, worth opening, and decoys, which cost more to open than they return. It has a cheap structural test χ — a shape predicate, priced by its depth — which is *evidence* rather than a decision procedure: rich specimens usually pass it, decoys sometimes do. It has a limited budget. If it opens too many decoys it dies.

Give it a crisp guard, and it opens whenever χ holds. In a generous world (rich specimens common, the test informative) this works well: the crisp forager achieves the highest terminal wealth of any strategy measured — and the lowest survival probability, 0.8501. In a harsh world, where the base rate of rich specimens drops to 0.15 and the test gets noisier, the crisp forager’s survival probability is 0.0029. It is, for practical purposes, extinct.

The graded forager survives at 0.9485 in the first world and persists in the second. The reason is not that it is cleverer. The reason is structural:

Slogan 0.5.1. *A Boolean guard has no coordinate in which to record disappointment.*

The crisp forager’s guard is either satisfied or not. When it opens a decoy, that outcome has nowhere to go. The formula χ was true; it is still true; nothing about the guard has changed. To learn from the disappointment the crisp learner would have to change the *formula* — move to a different, presumably deeper and more expensive test — which is a discrete jump, and an expensive one. The graded forager’s guard is the same formula, with a value attached, and the value moves: over five openings, its belief slides from strength 0.750 down to 0.250, and the probability that it opens the next specimen falls from 0.333 to 0.231. It has learned to be careful without learning anything new.

This is the sense in which graded where-clauses are a learning instrument. The formula language gives a learner a discrete space of hypotheses, and that space is a tree: to move from one hypothesis to a nearby one you must back up and branch, and there is a precise theorem [7] saying that a learner making only small moves in that tree can never leave the region it started in. Grading adds a continuous coordinate *inside* each region. The picture that results is not a space but a fibration: a tree of formulae, with a value-space hanging over each point. Search in the tree is backtracking; search in the fibre is something much closer to a gradient. A learner needs both, and until now the framework had only one.

0.6 Seeing is metered

The last ingredient is that observation costs something.

If the structural predicates are the learner’s *senses* — and they are; that is what it means for a formula to describe the shape of a term — then using them is work, and the work should be charged. The natural price is depth: a predicate that looks one constructor deep costs less than one that looks five deep, and a modality that requires actually driving the system a step costs more than a predicate that merely reads structure already present. In the underlying calculus this is not an add-on; there is a cost-accounting layer in which tokens are conserved, computation is funded, and a computation that runs out of budget stops [7, 13].

Once observation is metered, sentences one could not previously state become measurable. Here are three, all measured rather than argued.

Truth is not the best buy. In a study of noughts-and-crosses players whose strategies are rungs on a ladder of increasingly deep formulae [8], the player using the exactly-correct characteristic formula — the one that plays perfectly — is *dominated*. It wins 87.3% of games at a metered cost of 679 units, while a shallower heuristic wins 91.8% at 256. The correct theory is the only rung that never loses; it buys safety, not food.

Perception is cheaper than interaction, and it is where the yield is. On the same ladder, the three rungs that read structure the world has already deposited — with no modality at all — deliver 92.3% of the total gain for 34.3% of the total cost. The two rungs that require driving the system forward deliver the remaining 7.7% for the other 65.7%. There is a general thesis here about environments rather than learners: a predicate can be *read* if the environment has already computed it into legible structure, or *driven* if it has not, and the interesting question is which party is billed.

Whether truth is affordable is a property of the world. Run the same ladder construction on a different game — Nim [9] — and the result inverts. There the perfect strategy is the *cheapest* rung on the ladder and the best buy on it. Nothing about the learner changed. What changed was the environment’s structural regularity.

0.7 The range, in one page

Six disciplines are worked in Part VI. Each is a different setting of the dials, and each comes with a measured example.

1. **Spatial, crisp.** Only structural predicates, Boolean values. The senses. Worked on board games: forks, threats, and the discovery that requiring a fork to be expressible forces the encoding of the board.
2. **Modal, crisp.** Add the context-labelled modality: lookahead. Worked on the same games, where it turns out to be the most expensive and least productive part of the ladder.
3. **Graded, real-valued.** Values in $[0,1]$ or $\mathbb{R}_{\geq 0}$, supplied by a probabilistic logic. The mortal forager, and the discovery that the personality parameter governing scepticism has an interior metabolic optimum — credulous foragers open decoys and die, sceptical ones starve, and the optimum rises roughly sevenfold with the risk of the world.
4. **Graded, complex-valued.** Amplitudes rather than probabilities. Interference becomes available, which means Shor’s algorithm can be written in the guard language — and, more informatively, the semantics turns out to be *too* powerful rather than too weak, so that a discipline restricting clauses to be norm-preserving is what makes it a quantum language rather than a magic one.
5. **Semitopological.** The container is the opens of a semitopology, and the logic is about coalitions. Worked twice: on a hunting pride reading which way a herd will break, and on a honeybee swarm choosing a nest site. From the swarm comes the cleanest prediction in the corpus — that two rival quorums require twice the quorum threshold in scouts, so that a swarm smaller than that cannot split no matter what, whatever the site qualities or the inhibition dynamics.
6. **Resolution-graded.** The observation discipline is coarsened deliberately, and one asks what an observer at that coarseness reports. It reports fewer kinds than the world contains, and by a computable factor: at the coarsest resolution, one kind where there are four, and a population count inflated by $3.4\times$.

That last item is where the document ends, because it is where the argument becomes epistemology rather than engineering. Choosing an observation discipline is choosing which distinctions exist *for you*. A learner does not fail to see the distinctions it cannot afford; it fails to see that they are there, and reports a world with fewer kinds in it than the world has. That is a Kantian thought with a ledger attached, and the ledger is the contribution.

0.8 The evidence, on one page

Almost everything asserted above has been measured. Table 1 collects the headline figures; Part VI works each of them, and the simulations that produced them ship with the notes cited in the last column. A reader who stops at the end of Part 0 should at least have seen this table.

0.9 What is claimed, and what is not

Claimed. That the structural and behavioural components of a logic over a term language are determined by the presentation of that language, with no design freedom; that the remaining freedom is concentrated in two parameters, the collection container and the value algebra; that several established logics — intuitionistic, classical linear, coalitional — are settings of those

parameters; that placing such a formula in a binder’s guard makes it the resolution mechanism for the system’s own non-determinism; that grading the guard gives a resource-bounded learner a continuous coordinate it otherwise lacks; and that all of this is priced, so that the resulting claims about what a learner can afford to see are measurable and have been measured.

Not claimed. That every generated logic is well-behaved: adequacy — the property that formulae separate exactly what interaction separates — depends on the target logic having enough structure, and many useful settings do not have it. That the generated logic is decidable: it generally is not, and the practical discipline is to restrict either the language fragment or the connective set until model-checking terminates, at the cost of adequacy. That the results about foragers, prides and swarms are established biology: the swarm predictions in particular are untested proposals, offered because they are falsifiable and cheap to test, not because they have been confirmed. That this document proves much: it is a summary, and where it advances a new claim it is normally left as a conjecture, marked as such.

One methodological note. Almost every number quoted in this document comes from a simulation written alongside the note it appears in, and those simulations are in the same repository as the notes. Where a result is measured rather than proved, it is said to be measured. Where a result is a negative result — something the framework predicted that did not survive contact with the arithmetic — it is kept. There are several.

Part I — Generation

This part establishes the generator. A presentation of a computational system — a signature of term constructors, a set of rewrite rules, and a choice of how witnesses are collected — determines a logic, and the determination is mechanical for two of the three components. A fourth parameter, the algebra in which observations take their values, is introduced and shown to be independent of the other three. The part closes with a table exhibiting the logics used throughout this document as settings of the four parameters. The material of §1–§4 is a compression of [1]; the fourth dial is from [2, 12]; the adequacy discussion is from [13].

1 Realizability, and why the seams matter

Most accounts take a logic as given: a fixed vocabulary of connectives, a notion of model, and a truth predicate floating above computation, with truth values arriving from a Boolean or Heyting algebra that is postulated rather than built. For a discipline that wants its reasoning systems to be inspectable — traceable to constructions whose behaviour can be run and audited — this is the wrong starting point, because the glue binding truth to the algebra is opaque. Once it is in place one can no longer see what the logic is *made of*.

The commitment that makes the seams visible is realizability [30, 31]: a formula denotes the collection of computations that witness its computational content. Under that commitment, and one further fixing — of *how* witnesses are aggregated — a logic factors into three components, and every connective is of one of three kinds.

There is a further reason to care, beyond hygiene. If the connectives are operations of three named kinds, then a new logic is obtained by turning dials rather than by invention, and an algorithm defined over one logic can be transported to another by transporting the dial settings. That is the difference between a logic and a *procedure for generating logics*, and it is the practical content of this document. It also bears directly on a claim the AI community makes on its own behalf: that important aspects of intelligence can be faithfully rendered as computation. Logic has long been cast as the calculus of reason. Exhibiting logic itself as *arising from* computation is direct support for that claim, since the calculus of reason is thereby shown to be, at bottom, computational.

Notation 1.1. Throughout, T is the type of terms of the system under study, S is the collection type in which witnesses are aggregated, and $\llbracket \varphi \rrbracket : S(T)$ is the denotation of φ . We write $e \Vdash \varphi$ for “ e witnesses φ ”.

2 The three components

Structural connectives. Each term constructor lifts to a predicate transformer. If $f : T^n \rightarrow T$ is an n -ary constructor, define

$$\llbracket \widehat{f}(\varphi_1, \dots, \varphi_n) \rrbracket = \{ f(e_1, \dots, e_n) : e_i \Vdash \varphi_i \}.$$

The lifted constructor $\widehat{f}$ speaks about the *shape* of a witness: it asserts that the witnessing computation was built by f from sub-witnesses of the φ_i . For the rho calculus these are the

connectives of namespace logic [15, 16]: a connective for output, one for input, the separating conjunction $\varphi \mid \psi$ coming from parallel composition, and --- distinctively, and because the calculus is reflective --- the name predicate $@\varphi$, which holds of a name whose quoted process satisfies φ . Names see into structure.

Behavioural connectives. Each rewrite is labelled by the context in which it fires. A step $K[l] \rightarrow K[r]$ carries the one-hole context K , and lifting labels to modalities gives

$$\llbracket \langle K \rangle \varphi \rrbracket = \{ e : e \xrightarrow{K} e', e' \Vdash \varphi \}, \quad \llbracket [K] \varphi \rrbracket = \{ e : \forall e'. e \xrightarrow{K} e' \Rightarrow e' \Vdash \varphi \},$$

the context-indexed Hennessy–Milner modalities. There is one primitive modality per rewrite rule *and choice of redex position*; the quantifiers $\exists j. \langle K_j \rangle \varphi$ and $\forall j. [K_j] \varphi$ range over redex positions.

The indexing is not bookkeeping. In the noughts-and-crosses development [8] every move is the same action --- a square is claimed --- so an action-labelled logic sees one label and cannot tell nine moves apart, while the context-labelled logic sees as many labels as there are empty squares, and the branching factor of $\exists j$ is the legal move count. Getting this right required a correction: the redex context must be instantiated, which forced the encoding to make an empty square a receipt offering to be claimed, so that a ply is exactly one rewrite.

Remark 2.1 (The two halves of OSLF). Structural and behavioural connectives are the introduction and elimination halves of an Operational-Semantics-in-Logical-Form presentation. They are generated, respectively, from the constructors and from the labelled transitions, and neither involves any choice once the presentation is fixed. This is the precise content of the claim that the logics here are not *ad hoc*.

Collection connectives. Any operation on S lifts to a connective. With $S = \mathbf{P}_{\text{fin}}$ the semi-lattice operations give the familiar

$$\llbracket \varphi \wedge \psi \rrbracket = \llbracket \varphi \rrbracket \cap \llbracket \psi \rrbracket, \quad \llbracket \varphi \vee \psi \rrbracket = \llbracket \varphi \rrbracket \cup \llbracket \psi \rrbracket,$$

but this is exactly where the choice of container is felt, and Part II is about what varying it does.

Principle 2.2 (Where the freedom is). *Of the three components, two are determined by the presentation and one is a parameter. Design freedom in a generated logic is concentrated in the collection component --- and, once the fourth dial of §4 is added, in the value algebra.*

3 What the rewrite rules see, and what adequacy costs

Two consequences of Remark 2.1 should be recorded before they cause trouble later.

The rules determine the resolution of the modalities. The generated behavioural layer sees exactly what the rewrite relation distinguishes. If the rules are presented as interleaving, the logic sees interleaving; if they are presented so that independent steps commute as a genuine concurrency, the logic sees true concurrency. There is no separate decision to be made about which semantics the logic has. The presentation already made it.

Adequacy is a property of the target, not of the generator. It is tempting to say that a generated logic is automatically adequate — that its formulae separate precisely what interaction separates. This is false in general and the error is worth naming, since an earlier statement of this programme made it. Adequacy requires the target logic specification to supply enough structure, in particular something like both a conjunction and a negation. A generator can be asked to emit a logic without negation — there are excellent reasons to, discussed in Part IV — and the resulting logic will not be adequate. The Hypercube family of generators, which emits type systems rather than modal logics, offers no adequacy theorem at all, and type systems are not generally expected to be adequate [13].

This is not a defect but a dial of its own, and it is the dial a practitioner reaches for first. A user who wants model-checking to terminate gets it by restricting the fragment of the system, or the set of admitted connectives, or both. Restricting the system’s fragment *necessarily* restricts the structural and behavioural connectives, since those are generated from it. When model-checking terminates, the logic is necessarily not adequate for the whole of a Turing-complete system. The methodological posture throughout is therefore: *the ideal logic defines; restricted fragments decide*. Where an infinitary check is unavoidable, it is bounded by a token budget rather than forbidden — which, as Part IV shows, turns out to be the same budget that funds the learner’s observations.

Remark 3.1 (A caveat about the structural layer). The structural connectives are strictly more discriminating than context-labelled bisimulation: the spatial layer can separate terms that no interaction separates. This looks like a contradiction with adequacy and is not. For interactive systems there is a uniform extension of the presentation — adjoin an atom for each constructor, overload the interaction constructor so that applying it to that atom rebuilds the term, and add the corresponding destructors — which makes every structural distinction visible to bisimulation, at which point the two notions agree. The construction is folklore in the spatial-logic community, following Caires’ observation of the phenomenon [24, 23], and is made explicit in a companion note. We use the fact and do not reprove it.

4 The fourth dial: what an observation returns

The three components above answer *which* connectives a logic has. They do not answer what a formula, applied to a term, returns. The default answer is a truth value in $\mathbb{B} = \{\perp, \top\}$, and it is so thoroughly the default that it is rarely recorded as a choice.

Record it. Let V be an algebra of values, and let $\llbracket \varphi \rrbracket(P) \in V$. For the positive fragment, a commutative semiring suffices: conjunction is multiplication, disjunction is addition, and the constants $0, 1$ are the units. The instances used in this document are collected in Table 2.

Three observations about this dial, each developed later.

It is independent of the other three. The connectives available are fixed by (Σ, R, S) ; what they return is fixed by V . One can run the same spatial-behavioural formula over $\mathbb{B}$, over $\mathbb{R}_{\geq 0}$ and over $\mathbb{C}$ and get, respectively, a model-checking question, a propensity, and an amplitude.

It is where negation goes to die. The positive fragment needs only a semiring, but negation needs more. On $[0, 1]$ the map $1 - x$ serves; on $\mathbb{C}$ there is no canonical candidate and De Morgan fails. The design adopted in [2] is therefore two-sorted: a *crisp* sort with full Boolean structure

and fixed points, and a *graded* sort with semiring structure and neither negation nor fixed points. Part II explains *why* that is the right line to draw, by identifying the value algebras in which negation is available.

It changes what the logic is for. Over $\mathbb{B}$ a formula is a question. Over $\mathbb{R}_{\geq 0}$ it is a rate. Over $\mathbb{C}$ it is a contribution to a sum over histories, and disjunction acquires the possibility of being false with both disjuncts true — which is to say that interference is not an exotic addition to the logic but the ordinary behaviour of $\oplus$ once additive inverses exist.

5 In what sense the logics are not *ad hoc*

It is worth being exact about the claim, since it is easy to overstate.

What is claimed. Given a presentation (Σ, R) and a choice of (S, V) , the connectives of the logic and their semantics are determined. There is no step at which a designer says “and let us also have a connective for ...”. Two people who fix the same four parameters write down the same logic.

What is not claimed. That the four parameters are themselves forced. They are not; choosing them is the whole art. Nor that the resulting logic is *good* — adequate, decidable, or well-behaved. Nor that every logic in the literature arises this way. The claim is about the map from parameters to logics, not about its surjectivity.

Why this is more than tidiness. Three dividends recur.

First, *transport*. An algorithm written against the generated logic of one system transports to another by transporting the parameters. The knot classifier, the finite-geometry encoding, the board-game ladder and the swarm model are the same construction at four settings.

Second, *prediction*. Because the connectives are forced, one can predict which connectives a given setting will and will not have, and be surprised when a system in the wild turns out to have exactly those. The semitopology case of Part II is the sharpest instance: the framework predicts, from closure properties of a container alone, that the algebraic dual of a semitopology cannot have a meet and must instead carry a symmetric compatibility relation — which is exactly the structure Gabbay arrived at independently [34].

Third, *accountability*. When the logic is wrong for a purpose, one can say which dial was set wrong, rather than starting over.

6 The instantiation table

Table 3 exhibits the logics used in this document as settings of the four parameters. It is the central claim of Part I in one object, and the rest of the document is an unpacking of its rows.

Two rows of that table deserve immediate comment, because they are the ones a reader is most likely to think cannot be right.

The linear logic row is worked in §9. The claim is not that linear logic can be *modelled* in this framework — of course it can, it is a logic — but that it is what the generator emits at a particular container setting, and that its two conjunctions, its exponential and its involutive negation each correspond to an identifiable feature of that container.

The coalition row is the one that comes with independent corroboration. It is also the row that discharges a promise: [1] treats only the collection component of coalition logic and defers the structural and behavioural components, together with the identification of the underlying points with ρ terms, to a companion. Part II, §10, is that companion.

Part II — Four containers, four logics

This part works the collection dial. It shows first that the algebraic theory presenting the container is the substructural hierarchy of the induced propositional logic; then that a second, finer axis is needed — which lattice operations the container makes primitive — and that three settings of that axis give intuitionistic logic, classical linear logic, and the logic of decentralised consensus. Linear logic is worked in detail, because it is the setting at which the framework’s claim to be generating rather than modelling is most testable: its two conjunctions, its exponential and its involutive negation each turn out to name a feature of the container. The part closes by supplying the structural and behavioural components for the coalition-logic row, which [1] deferred to a companion. Sources: [1] throughout; [17, 18, 19] for the linear row; [33, 34, 5, 6] for the semitopological row.

7 The substructural dial

The collection type S is a monad, and the algebraic theory presenting it is not incidental: *its equations are the structural rules of the propositional logic the collection component induces*. Write the binary aggregation as $*$ with unit ε . Then

commutativity of $*$	$\longleftrightarrow$	exchange,
idempotence of $*$	$\longleftrightarrow$	contraction,
unit and absorption laws	$\longleftrightarrow$	weakening.

Reading the standard containers through this correspondence:

- P_{fin} , finite powerset — a commutative idempotent monoid, i.e. a join-semilattice — gives exchange and contraction, and with the unit, weakening: the full structural regime, hence Heyting or Boolean propositional logic.
- **Bag**, finite multisets — the free commutative monoid, commutative but not idempotent — gives exchange without contraction: the multiplicative-linear regime, with $\otimes$.
- **List** — the free monoid, neither commutative nor idempotent — gives neither: a non-commutative, Lambek-style regime [20].
- **D**, distributions — commutative, with a convex structure in place of idempotence — gives the probabilistic regime.

Principle 7.1 (Collection is substructure). *The setting of the collection dial is the pair of Booleans (commutative?, idempotent?) of S , and that pair is exactly the presence or absence of exchange and contraction in the induced logic.*

Two coordinates, and everything in this part is bookkeeping on them. It is worth noticing already that the same two coordinates are the ones on which the no-go theorems for distributive laws between monads turn [38]: the obstruction to composing two containers and the price of packaging a container’s own sub-collections back into itself are one datum read twice.

Remark 7.2 (Commutative only). The non-commutative row is real and we do not develop it. The reason is that the system this document is anchored in has an associative-commutative parallel composition, so its parands form a bag, not a sequence; the natural container is **Bag** and the natural linear logic is the commutative one. Where genuine sequence structure appears — as it did when the weighted-rewriting implementation had to add an ordered term former, and with it the corresponding positional structural connective, to stop two distinct synapses canonicalising to the same channel [12] — the **List** row is what one would want, and Lambek calculus is where it leads. That is a different paper.

8 A finer axis: which lattice operations are primitive

Principle 7.1 is not the end of the story, because two containers can agree on both Booleans and still induce very different logics. The finer question is *which lattice operations the container makes primitive*, as against merely definable or merely relational.

Consider three containers whose aggregation is union — commutative and idempotent in every case, so indistinguishable on the dial of §7 — and yet:

- The opens of a *topology* form a frame: arbitrary joins and finite meets are both primitive, and the induced logic is geometric, hence intuitionistic. A proposition is an affirmation: something that can be positively *said* [32].
- The opens of a *semitopology* [33] — closed under arbitrary unions but *not* under intersection — form a semiframe: only arbitrary joins are primitive. Conjunction survives relationally and derivatively but not as a container operation. The induced logic is about what can be positively *done*.
- A quantale of the form $\mathbf{P}_{\text{fin}}(M)$ for a monoid M has arbitrary joins *and* a multiplication inherited from M — and, if the lattice is closed under intersection, a second, independent conjunction. Two conjunctions, from two different levels of the container.

Principle 8.1 (Primitivity). *Beyond which structural rules hold (Principle 7.1), a container is graded by which lattice operations it makes primitive. A frame makes arbitrary $\bigvee$ and finite $\bigwedge$ primitive. A semiframe makes only $\bigvee$ primitive. A quantale makes $\bigvee$ primitive and adds a multiplication that need not agree with any meet.*

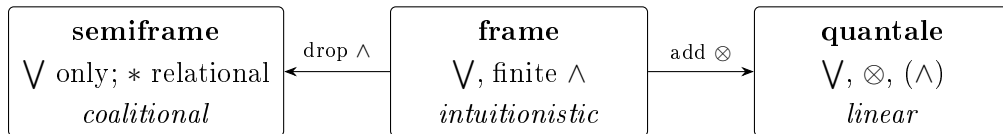


Figure 2: Three settings of the primitivity axis. The middle is the familiar case; the logics of interest here are on either side of it.

The rest of this part works the two ends of Figure 2.

9 Linear logic, generated

In plain terms: linear logic is what you get when witnesses are collected in a container that keeps count. Its notorious pair of conjunctions is not a quirk of the proof theory — it is the signature of a container with two layers, one that multiplies and one that intersects.

9.1 Phase spaces are a two-layer container

Girard’s phase semantics [17] fixes a commutative monoid M and a distinguished subset $\perp \subseteq M$, the *pole*. For $X \subseteq M$ put

$$X^\perp = \{y \in M : \forall x \in X. xy \in \perp\},$$

and call X a *fact* when $X = X^{\perp\perp}$. The connectives are then

$$X \otimes Y = (X \bullet Y)^{\perp\perp}, \quad X \bullet Y = \{xy : x \in X, y \in Y\}, \quad X \& Y = X \cap Y, \quad X \oplus Y = (X \cup Y)^{\perp\perp},$$

with $X^\perp$ the linear negation, $\wp$ and $\multimap$ recovered by duality.

Read this through §2. The container is not one thing but a stack:

$$\underbrace{\mathbf{Bag} A}_{\text{inner: a monoid}} \longrightarrow \underbrace{\mathbf{P}_{\text{fin}}(\mathbf{Bag} A)}_{\text{outer: a lattice}} \longrightarrow \underbrace{\text{facts}}_{\text{the nucleus}}$$

The inner layer is the free commutative monoid on the atoms — a bag. It carries a multiplication and nothing else. The outer layer is the powerset, which carries arbitrary unions and intersections. And the pole induces a closure operator $X \mapsto X^{\perp\perp}$ — a nucleus — which cuts the outer lattice down to the facts and, in doing so, makes negation involutive.

Observation 9.1 (The two conjunctions are two layers). $\otimes$ is the inner monoid lifted pointwise and then closed; $\&$ is the outer container’s intersection. They are different connectives because they come from different levels of the container stack, and this is why they obey different structural rules: the inner multiplication cannot discard or duplicate a factor, because the bag records multiplicity, while the outer intersection is set-shaped and therefore fully structural.

Observation 9.1 is the whole of the “linear logic is generated” claim in one line, and it does real explanatory work. The multiplicative / additive split is presented in most introductions as a discovery about sequent calculus — two ways to combine contexts, one sharing and one splitting. Here it is the difference between the two floors of a container, and the sequent-calculus fact follows rather than being the primitive.

Remark 9.2 (Quantales). Abstractly, $\mathbf{P}_{\text{fin}}(M)$ is the free quantale on the monoid M : a complete lattice with an associative multiplication distributing over arbitrary joins [19]. A quantale equipped with a dualising element is a *Girard quantale*, and Girard quantales are exactly the algebraic models of classical linear logic [18]. Everything in this section can be run in that generality; the phase-space presentation is the concrete case $M = \mathbf{Bag} A$, which is the one our term structure hands us.

9.2 The separating conjunction is the multiplicative

Now specialise the atoms to terms. In the ρ calculus parallel composition is associative and commutative with unit 0, so the parands of a term form a bag and $(T, |, 0)$ is a commutative monoid. Lift it pointwise. What is obtained is

$$\llbracket \varphi \mid \psi \rrbracket = \{ P \mid Q : P \Vdash \varphi, Q \Vdash \psi \},$$

which is precisely the *separating conjunction* of spatial logic [23, 16] — the connective that says a term splits into a part satisfying φ and a part satisfying ψ .

Observation 9.3 (One connective, two names). *The structural connective generated by parallel composition and the multiplicative conjunction of the induced Girard quantale are the same operation: the container’s monoid lifted pointwise. Spatial logic’s $\mid$ is linear logic’s $\otimes$. Correspondingly, the graded separation $\varphi \mid_k^N \psi$ used throughout the mortal-scientist work — which permits the two parts to share at most k names of a described namespace N — is a graded $\otimes$, and $k = 0$ is the strict, resource-disjoint case.*

This is folklore-adjacent and should be positioned honestly. The logic of bunched implications [21] and separation logic [22] make the same move over a *partial* commutative monoid of heaps, where two heaps compose only if their domains are disjoint, and where the additive conjunction sits alongside the multiplicative rather than being derived from a second container layer. The difference in our setting is instructive: the ρ monoid is *total*. Any two terms compose. So the splitting witnessing $\varphi \mid \psi$ is not unique and not determined by a disjointness side condition; it is chosen, and there may be many.

That non-uniqueness is not a nuisance. It is the point of contact with the construction [1] is organised around. Membership in $X \bullet Y$ is an *existential over splittings* of an element, and the object of splittings of a term is the pair $(\partial T, T)$ — a one-hole context together with what sits in the hole — which is the McBride derivative [28, 29], the zipper, the *proto power object*. Assembling the splittings back into a collection of the same shape as the container is the operation [1] calls packaging, and its central result is that packaging is a section of the quotient presenting the container, whose choice content is exactly the assertion that the atoms admit a linear order.

Observation 9.4 (The multiplicative is where the packaging problem lives). *$\otimes$ is definable relationally without any choice — to know whether P satisfies $\varphi \mid \psi$ one need only know that some splitting works. To produce the collection of witnessing splittings as an object of the container’s own shape requires packaging, and hence choice. The relational form is free; the operational form is not. Linear logic’s multiplicative is therefore the natural worked instance of the proto-versus-packaged distinction, and not a separate topic.*

Observation 9.4 will be picked up again in Part V, where the same distinction reappears in a completely different costume: a graded where-clause *weighs* the candidate resolutions of a race without selecting one, and a scheduler *selects*. Weighing is relational and free; selecting is packaging and is priced.

9.3 The exponential

The remaining connective is the exponential $!$, and on the reading above it has an unusually concrete description: $!$ is the modality that re-licenses the structural rules the container withheld. Contraction corresponds to idempotence (Principle 7.1); the container was chosen to be a bag

precisely so as *not* to be idempotent; and $!\varphi$ marks the propositions on which one may nonetheless duplicate. In container terms it is the passage from **Bag** back up to P_{fin} , restricted to those elements where the passage is legitimate.

Remark 9.5 (What the exponential costs). There is a price attached, and [1] computes it. Packaging for **Bag** needs a linear order on atoms, so that each finite support has a canonical sorted representative. Packaging for P_{fin} needs that *and* a deduplication step, hence decidable equality on atoms. The additional equation between the two containers is idempotence, and the additional resource is an equality test. So: the modality that restores contraction is exactly the packaging cost of idempotence, and that cost is a decidable equality. In a reflective calculus where atoms are quoted processes, “decidable equality on atoms” is a substantive and recognisable requirement.

Remark 9.6 (The exponential in the calculus, and a conjecture). It is tempting to identify ! with a syntactic feature of the calculus, and the temptation is worth recording without being indulged. Rholang receipts come in two flavours: linear, written $<-$, which is consumed when it fires, and persistent, written $<=$, which is not. A persistent receipt is precisely a capability to receive that may be used any number of times, which is the informal gloss of ! on the input side; and the trophic vocabulary of the mortal-scientist work makes the same distinction for the other side of a communication, separating a *source* — persistent, rate-limited, non-exclusive — from *prey* — a linear send, exhaustible, exclusive [7]. Same sort, same rewrite rule, different multiplicity.

There is also a suggestive piece of independent evidence. In the graded setting, what makes a race between two messages a race at all is the *linearity* of the receipt: the loser’s message survives and constitutes a which-path record, so the two branches cannot recombine. Replace the receipt by a persistent one and both messages are eventually consumed, the receipt survives in either branch, and the derivations reconverge — the interference site is destroyed [2]. The exponential erases which-path information. That is exactly what one would want ! to do to a resource-sensitive semantics, and it is checkable rather than metaphorical.

We nevertheless leave the identification as a conjecture rather than promoting it to a proposition. Making it precise requires a statement about contraction on received data in the presence of reflection, which is a technical question with a known trap in it: reflection is the mechanism by which a term can acquire structure it did not syntactically contain, and it is already the suspect in several open questions elsewhere in this corpus. The conjecture is that persistence is the calculus’s native !; the honest status is that we have the resource reading, the cost reading, and one behavioural test, and not a proof.

9.4 Why this row matters for the rest of the document

Two dividends, both cashed later.

First, the value dial and the collection dial turn out to be the same dial applied at different layers. A quantale is a container of truth values; grading a formula in a quantale, as [14] does for the **rho** calculus, means its truth values are linear-logic truth values, with $\otimes$ for conjunction and $\vee$ for disjunction. This is not a pun: it is the observation that the algebra one uses to *collect witnesses* and the algebra one uses to *value formulae* are subject to the same constraints, because in both cases the connectives are the operations under which the algebra is closed.

Second — and this is the payoff advertised in Part I — the linear reading supplies a criterion for something the graded-clause design had only by inspection.

Observation 9.7 (When the graded sort has a negation). *The graded sort of [2] was given a semiring structure and denied both negation and fixed points, on the grounds that $1 - x$ works on $[0, 1]$ but $\mathbb{C}$ has no canonical negation and De Morgan fails. The quantale reading explains the line. A quantale supports an involutive negation exactly when it carries a dualising element, i.e. when it is a Girard quantale [18, 19]. Then:*

- $[0, 1]$ with Lukasiewicz negation $1 - x$ is a Girard quantale — which is why probabilistic-logic values support refutation as $1 - s$, and why the mortal forager’s confirm / refute / undetermined trichotomy sits as three corners of a square rather than requiring a separate negation in the hypothesis language;
- a general quantale supports fixed points by Knaster–Tarski, because it is a complete lattice — which is why the graded fixed point is available in quantale-graded ρ and has to be argued for separately over $\mathbb{R}_{\geq 0}$, where it becomes a convergence condition;
- $\mathbb{C}$ is a rig with no order at all, hence not a quantale, hence carries neither a dualising element nor Knaster–Tarski. Negation and fixed points are both unavailable, and for the same reason.

So the two-sorted design is not a compromise forced by the complex case. It is the statement that a crisp sort is needed exactly when the value algebra fails to be a Girard quantale.

Observation 9.7 is, as far as we know, new, though it is a short step from facts that have been available since [18]. It is recorded here rather than in Part IV because it is a fact about containers, and it is the reason Part IV can treat negation as a single phenomenon with several faces rather than as a recurring nuisance.

10 The semitopological row, and the promised companion

In plain terms: if the container is the set of coalitions that can act on their own, then two coalitions that can each act may overlap in a way that cannot. Conjunction stops being an operation. What is left is a logic of what can be positively done, and it is the logic of decentralised consensus.

10.1 The collection component

A *semitopology* [33] is a pair (P, Open) with $\emptyset, P \in \text{Open}$ and Open closed under arbitrary unions, but not necessarily under intersection. Points are participants; an open is an *actionable coalition*, a set of participants with the resources to act without anyone else’s permission. A union of actionable coalitions is actionable; an intersection need not be. That one dropped axiom is the whole difference from topology.

Take $S = \text{Open}(P)$ and run §2. What comes out, with nothing added by hand, is: arbitrary disjunction is a genuine collection connective; conjunction is *not* interpretable as a container operation, though the interior $\text{int}(A \cap B)$ exists as a derived operation by union-closure; and the primitive relational residue of conjunction is the compatibility relation

$$A * B \iff A \cap B \neq \emptyset,$$

recording co-realizability — the existence of a witness realising both — without producing a collection of witnesses.

Observation 10.1 (The framework predicts the semiframe). *From the closure properties of the container alone, the collection-component analysis predicts that the algebraic dual of a semitopology is a complete join-semilattice equipped not with a meet but with a symmetric, join-distributive compatibility relation. That is exactly Gabbay’s semiframe [34], arrived at independently as the structure needed to make the semitopology–semiframe duality work and to express the intertwinedness conditions governing consensus. Two routes, one relation.*

It is worth recording why this is the best available evidence for the whole programme. The prediction was not tuned: the container analysis was developed for reasons having nothing to do with distributed systems, and the structure it predicts is idiosyncratic — a relation where one expects an operation, with three Horn-shaped axioms. Finding it already in the literature, derived differently, is the kind of corroboration a generative account is supposed to produce and a descriptive one is not.

Remark 10.2 (Compatibility is a proto-conjunction). The parallel with §9 is exact and worth one line. The relation $*$ is to conjunction what the object of splittings is to the power object: the un-packaged, relational form that records possibility without producing the packaged object. Recovering an actual conjunction from $*$ means applying an interior — a closure, a global operation — just as recovering a sub-collection from splittings meant applying `pack`.

10.2 Points are processes: the structural and behavioural components

[1] treats only the collection component of this row and defers the rest. We supply it, following [5].

The starting point is a puzzle in the consensus-types development [6]. Its participants are channel names, an open set is a set of channels, and the coalition modality $\exists\Box\varphi$ at a participant is realised by a *join* — a receive that must synchronise on several channels at once. The paper observes that the coalition modality decomposes over *participants* rather than over *process structure*, and flags a systematic comparison with spatial logic as future work.

Proposition 10.3 (The distinction collapses under reflection). *In the rho calculus the two decompositions coincide. Participants are processes; the channel names by which they are addressed are quoted processes; and the name predicate $@\varphi$ lets a formula about process structure be asked of a name. Consequently an open set — a set of participants — is a described sub-namespace, and quantification over opens is quantification over namespaces described by name predicates.*

The proof is by inspection of the encoding and is given in [5]; what matters here is the consequence. The structural component of the coalition logic is namespace logic, and the behavioural component is the ordinary context-labelled modality $\langle K_j \rangle$. There is no third kind of connective. What distinguishes this row from the spatial-behavioural row is the container, and only the container.

Writing **Open**-indexed aggregation explicitly, the coalition modality over a namespace N and a value algebra V is

$$\llbracket \langle E \rangle_N^V \psi \rrbracket = \bigoplus_{\substack{O \subseteq N \\ O \text{ open}}} \bigotimes_{q \in O} \llbracket \psi \rrbracket(q),$$

— an $\oplus$ over coalitions of an $\otimes$ within one — and this single display is where the three dials meet. The container supplies the quantification pattern; the term language supplies what ψ can say about each participant; the value algebra supplies $\oplus$ and $\otimes$.

10.3 The container forces the connective; attainment forces the algebra

Given the display above, which V should one use? The question looks like a modelling decision and is not.

Over $\mathbb{R}_{\geq 0}$ with $\otimes = \times$, a coalition's value is a product of per-participant values, so a large coalition decays toward zero: bigger quorums would always be weaker, which inverts the intended semantics. The correct quorum reading is that a coalition is as strong as its weakest member, i.e. $\otimes = \min$ — the tropical and Viterbi algebras.

That would be a modelling preference, except that the same two algebras are picked out independently by a condition in Part V. A graded clause assigns a value to each candidate; a scheduler must actually select one; and the selection is *attained* — realised by a definable selection function returning an actual candidate rather than an expectation — exactly when $\oplus$ is idempotent, with max and min the two cases of interest [3]. So:

Observation 10.4 (Three constraints, one algebra). *The container forces the shape of the coalition connective, since a meet-free container can only offer $\oplus$ -over-opens of $\otimes$ -within. The intended quorum semantics forces $\otimes = \min$. Attainment forces $\oplus$ idempotent. The intersection of the three is the tropical/Viterbi row of Table 2 — which is precisely the row where a selection function exists on the nose, so that an observer evaluating the modality receives a witness coalition rather than a scalar.*

The ecological reading of that observation is the reason it is here rather than in an appendix. A lion pride reading which way a stampeding herd will break needs to know not merely how strong the consensus is but *which way*; a honeybee scout committing to a nest site needs a site, not a number. The framework lands, for independent reasons, in the one fragment where selection is definable — which is why an animal can act on it at all.

10.4 What changes when the observer is mortal

One further move separates this row from the consensus literature, and it is a change of direction rather than of content.

The consensus-types development reads the correspondence as a term assignment: from a proof of φ , synthesise a protocol P realising it. Its open problems are accordingly proof-theoretic — there is no proof system for coalition logic yet, and supplying one is most of the work. A mortal observer needs the opposite arrow, $P \models \varphi$: given a population, has a coalition formed? Same logic, opposite direction.

Proposition 10.5 (The observation reading is proof-theory-free). *Reading the coalition logic as a hypothesis language about an already-running population requires only its semantics, which the container supplies. It requires none of the proof-theoretic apparatus that synthesis requires.*

And with that reversal comes a reframing that Part VII will take up. Gabbay's semitopology is *given*: the modeller supplies the set of actionable coalitions. A mortal observer has to *discover* it. The semitopology is therefore not the model but the *hypothesis* — the thing the observer is guessing at — and coalition logic stops being a specification language and becomes a hypothesis language. That is the sense in which a container choice is also an epistemological commitment: it says what kind of thing the observer takes its world to be organised into.

Part III — Placement

Having a logic is not yet having an instrument. This part distinguishes three things one can do with a formula — synthesise a program from it, test a program against it, or install it in a program as a guard — and argues that the third placement is what turns an observation discipline into something a system can look through. It also introduces a fourth use, the formula as a revisable hypothesis, which is what Part VII will need. Sources: [7] for the assay and the hypothesis lattice, [5] for the reversal, [13, 2] for the guard syntax.

11 Three placements

Fix a generated logic $\mathcal{L}(\Sigma, R, S, V)$ and a formula φ of it. There are three essentially different things to do with φ , and they have different costs, different prerequisites and different users.

Specification: $\vdash \varphi \triangleright P$. From the formula, produce a program realising it. This is proofs-as-programs, and for the logics of Part II it is the direction with the most literature and the most open problems. It requires a proof system, and for the coalition row there is not yet one: the consensus-types development lists the extraction of a proof system as its central gap, and four of its seven open problems are downstream of that gap [6].

Test: $P \models \varphi$. Given a program, decide — or, within a budget, attempt to decide — whether the formula holds. This requires only the semantics, which the generator supplies with the connectives. This is what a model checker does, and what an *observer* does.

Guard: φ **inside the binder.** Install the formula in a for-comprehension:

$$\text{for}(ptrn_{11} \leftarrow chan_{11} \ \& \ \cdots \ ; \ \cdots \ ; \ ptrn_{1n} \leftarrow chan_{1n} \ \text{where } \varphi) P$$

where $\&$ joins within a group and $;$ separates groups, and where the preceding clauses bind the variables the condition may mention [13]. Now the formula is evaluated when a communication is about to occur, against the candidate data, and its value decides whether that communication occurs.

The three are related but not interchangeable, and the differences are the substance of this part.

12 The arrow reverses, and the cheap direction is the useful one

The clearest instance of the difference is the coalition row, where the same logic has been developed in both directions within this corpus.

The consensus-types development [6] reads the correspondence as a term assignment. Its table maps participant to channel name, open set to channel set, and the coalition modality at a participant to a *join*; its right rule introduces the join, its left rule introduces a send — “contribute local evidence” — cut is communication, and cyclic proof is guarded recursion. The computational content of the quorum-intersection lemma is that any two joins share a channel

carrying a well-typed message. It is a synthesis story: from a specification of a consensus property, build a protocol.

A mortal observer wants the opposite. A hunting pride does not want to build a herd; it wants to know which way the herd will break. Its question is $P \models \varphi$, not $\vdash \varphi \triangleright P$.

Proposition 12.1 (The observation reading sidesteps the proof theory). *Reading a generated logic as a hypothesis language about an already-running system requires its semantics only. The proof-theoretic apparatus that synthesis requires — and which, for the coalition logic, does not yet exist — is not on the critical path.*

This is the strongest strategic argument available for the whole observation-discipline programme, and it is worth stating in general rather than for one row. The generator hands you semantics for free, because the semantics *is* the lifting of constructors and contexts. It does not hand you a proof system; proof systems have to be built, and building one is the expensive part of adopting any new logic. An observer does not need one. So the range of logics that are practically available for observation is far larger than the range that is practically available for synthesis, and the dials of Part I can be turned freely in the first setting long before the second catches up.

Remark 12.2 (Where the guard sits). The guard placement is on the observation side of this divide, not the synthesis side. Evaluating φ against a candidate message is a satisfaction question, which is why a where-clause is cheap to add to a language and why it inherits whatever fragment discipline the model checker was already under.

13 The guard, and what it is for

In plain terms: a concurrent system has to decide, constantly, which of several possible communications happens next. Ordinarily that decision is made by the hardware. A guard moves it into the language, and makes it a question about what the system can observe.

A term of the form

$$x!(Q_1) \mid \text{for}(y \leftarrow x)P \mid x!(Q_2)$$

is a race. Two messages are available on one channel and one receipt is waiting; exactly one of them will be consumed, and nothing in the calculus says which. In a production implementation the answer is supplied by the physical race of actual cores against a shared store. That is a resolution mechanism, but it is one the language cannot speak about, cannot reason about, and cannot vary.

Installing a formula on the binder changes this. The guard is evaluated against each candidate, and its value determines whether — and, once graded, with what propensity — that candidate is taken. The non-determinism is still there, but it is now resolved by *something the system can say*.

Principle 13.1 (The guard is where resolution becomes legible). *A logic in the specification placement describes a system. A logic in the guard placement participates in it: the formula's value is the mechanism by which the system's own non-determinism is resolved. Everything the observation discipline can distinguish is thereby available to the scheduler, and nothing else is.*

Three consequences, each developed later.

Per-match evaluation. The clause is evaluated once per candidate match, not once per channel state. This is a design decision and it is the one that makes the rest work: without it there is no map from candidates to values, and hence nothing for Part V’s weighing-versus-selecting distinction to act on.

Receipts are linear by default. A linear receipt is consumed when it fires; a persistent one is not. The default matters more than it looks. Under a persistent receipt the term above is not a race at all: both messages are eventually consumed, the receipt survives either way, and the two derivations reconverge. It is *linearity* that makes the surviving message a record of which branch was taken, and therefore linearity on which the whole treatment of racing in Parts IV and V rests [2].

Perception must not be able to act. If a formula is to read structure without disturbing it, the reading must not itself be a communication that consumes something the system needed. This is a real design constraint and it bites in practice: in the board-game encoding, threat markers had to be given their own name family, because a line signalling a threat on a square’s own move channel would rendezvous with that square’s receipt and *play the move*. The same separation is forced in the herd model, where a scan implemented with an ordinary receive rather than a peek would mean that the pride moves the zebra by looking at it [8, 5]. Grades of access are separated by choice of namespace, not by good intentions.

14 The hypothesis placement

The fourth use of a formula is the one a learner makes: not to check a program once, but to hold a guess and revise it.

14.1 The assay

The mortal-scientist development [7] makes an experiment out of the guard placement directly. An *assay* of a hypothesis φ against a specimen arriving on channel p is a complementary pair of guarded receipts:

$$\text{for}(x \leftarrow p \text{ where } \varphi(x))\{\text{confirm}\} \quad | \quad \text{for}(x \leftarrow p \text{ where } \neg\varphi(x))\{\text{refute}\}$$

Exactly one of three things then happens: the confirming branch fires, the refuting branch fires, or neither does before the budget runs out. That trichotomy — confirm, refute, undetermined — is the note’s central epistemological object, and the third outcome is not a modelling convenience: for a mortal observer, *refutation is budget-relative*. A step not yet enabled is indistinguishable from a step never enabled, and telling them apart is exactly what a finite purse cannot buy.

Two features of this construction will be picked up repeatedly.

First, it needs negation *in the hypothesis logic*, and by §3 a generated logic supplies negation only if the target specification does. The assay therefore carries an unadvertised dependency, which Part IV discharges.

Second, the undetermined outcome has two sources that an observer cannot separate from the outside. In the consensus setting, one is a property of the participant — stuck, faulty, Byzantine — and the other is a property of the observer’s purse. A mortal scientist cannot distinguish a faulty participant from an underfunded observation. Measured in the herd model: with a purse

of 60, 61.04% of verdicts are undetermined, of which 56.68% would have been decidable with a richer purse; at a purse of 200 or more the undetermined outcome disappears entirely [5].

14.2 The lattice, and the shape of revision

Hypotheses are formulae, and formulae are ordered by logical strength, so the hypothesis space is a lattice — the *relaxation lattice*, obtained by weakening a characteristic formula along its structure. It also carries a metric: formulae agreeing to modal depth n are within 2^{-n} of each other, which is an ultrametric.

That geometry has consequences a designer should know before choosing a search strategy.

- The balls are nested or disjoint, every point is a centre, and there is no betweenness. *There is no gradient.* Gradient methods are not ill-suited here; they are ill-typed. What is available is tree search.
- A revision move is a pair (locus, operation); its distance is the depth at which one had to back up, and its direction is which sibling one moved to.
- Cost is monotone in distance: a move of distance 2^{-n} invalidates every confirmation obtained at depth $\geq n$. Backtracking is not free; it discards evidence.
- **Confinement.** A walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it started in. A purely incremental learner cannot escape its initial basin — not because it is not clever enough, but for metric reasons.

Confinement was measured, not merely proved. In the board-game study, the best opening *reverses* as theory depth increases: at one rung the centre opening wins .833 against the corner's .597, while under the exactly-correct theory the corner wins .896 against the centre's .548. Deepening the theory while holding the opening fixed therefore makes things worse, $.833 \rightarrow .548$. The corner-opening mutant invades at identical cost and is exactly the distance- 2^{-1} move that Confinement denies an individual [8].

14.3 Why this part matters for the next one

The picture just described has one degree of freedom and it is discrete. A learner holds a formula; to change its behaviour it must change the formula; changing the formula is a jump in a tree, priced by how far it had to back up, and small jumps cannot leave the neighbourhood.

Part IV adds a second degree of freedom that is continuous and lives *inside* each ball. The consequences for learning are taken up in Part VII, but the mechanism is entirely the subject of the next part: let the guard return a value rather than a bit.

Part IV — Grading

This part lets the guard return a value instead of a bit. The syntactic change is small; the consequences are not. Weight maps — the standard device for attaching rates to rewrite rules — turn out to be a normal form of graded clauses rather than an independent mechanism, which retires an awkward side-condition. State-dependence becomes free. Interference becomes available, and with it a syntactic criterion for when a race can interfere at all. And negation, which has been surfacing as an obstruction since Part I, is assembled into a single phenomenon with five faces. Sources: [2] throughout, [12] for weight maps, [14] for the quantale case, [43] for the probabilistic-logic instance.

15 The design

Let V be a value algebra from Table 2. A *graded where-clause* is a formula whose value at a candidate match lies in V :

$$\psi ::= [\beta] \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle_V \psi$$

where $[\beta]$ embeds a crisp formula of the generated logic, v ranges over constants of V , and $\langle K \rangle_V$ is the graded modality — a transfer operation summing, over the transitions available at K , the value of ψ at the target.

Three decisions fix the design.

Two sorts. The crisp sort keeps full Boolean structure and fixed points; the graded sort has neither. By Observation 9.7 this is not an ad hoc restriction but the statement that a crisp sort is needed exactly when V fails to be a Girard quantale. Where V is a Girard quantale — $[0, 1]$ with Łukasiewicz negation, or any quantale-graded setting [14] — the graded sort can carry negation and, by Knaster–Tarski, fixed points too, and the two sorts collapse into one. Where $V = \mathbb{C}$ they cannot, and the crisp sort earns its keep.

Per-match evaluation. The clause is evaluated once per candidate match. This is what makes the clause a *function on candidates*, which is the object everything downstream acts on.

The value is invisible to a classical reader except through its support. A classical interpreter reads the clause as a Boolean — fires iff the value is non-zero — and behaves exactly as before. This keeps the language un-forked: the same program means one thing under a classical resolver and another under a graded one, and the graded meaning refines rather than replaces.

Theorem 15.1 (Conservativity). *At $V = \mathbb{B}$ the graded design recovers the unmodified where-clause semantics exactly.*

Corollary 15.2 (Support adequacy). *For every term P , $\text{supp}[\![P]\!] \subseteq \text{Reach}(P)$: the graded semantics assigns non-zero value only to outcomes the operational semantics can actually reach.*

The inclusion is one-sided on purpose. Where it is *strict* — where an operationally reachable outcome carries value zero — that is the *interference deficit*, and it is the operational signature of destructive interference. A two-sided condition would forbid exactly the phenomenon the complex case exists to expose.

16 Weight maps are a normal form

The standard way to attach quantitative behaviour to a rewrite system is a *weight map*: each rule carries a table whose keys are spatial-behavioural formulae refining the rule’s left-hand side and whose values are weights, so that a redex is classified by which key it satisfies and weighted accordingly [12]. Since the successor state carries such a table, rewrites can update them, and weights are dynamic. This machinery supports Gillespie-style stochastic simulation over $\mathbb{R}_{\geq 0}$ and a quantum continuous-time chain over $\mathbb{C}$.

It also carries a well-known irritant. For the classification to be well defined the key set must be a *partition* of the refinements, which has to be declared, checked, and enforced — and enforcing it turned out to require a further requirement that keys be structural, since modal keys break the locality argument on which incremental recomputation of propensities depends. The implementation ended up funnelling every partition through a single checked constructor, and then had to make the relevant struct fields private after it was discovered that a struct literal could route around the check [12].

Theorem 16.1 (Weight maps are graded clauses in disjunctive normal form). *A weight map is a linear combination of crisp indicator formulae with constants from V . General graded clauses are strictly more expressive: overlapping conditions are permitted, and no partition is required.*

Corollary 16.2 (The partition discipline is unnecessary). *Single-valuedness is automatic, because a formula has one value. The partition existed only to make a table lookup single-valued.*

Proposition 16.3 (Plasticity is free). *A clause is evaluated against the current term, so a value that depends on the system’s state requires no update function and no separate mechanism. Later values depend on earlier firings automatically.*

Proposition 16.3 deserves a moment, because it is not merely an engineering simplification. In the weight-map presentation, state-dependence is a side-channel: an update function mapping a table to a new table, whose composition order matters and whose Markov property has to be argued for. In the clause presentation it is the ordinary behaviour of evaluation. Part V will show that this same freeness is, in the choice-theoretic reading, exactly the principle of dependent choice — obtained with no new syntax at all.

Remark 16.4 (Where the control belongs). There is an ergonomic claim underneath the formal one. A state-side table is a specification of a generator: it says what the system’s dynamics are. A clause on a binder is an *arranging instrument*: it says what this particular communication should prefer. A programmer trying to make interference do something useful is arranging, not specifying, and the control should be where the arranging happens.

17 What the graded reading buys, and what it costs

17.1 Sum over histories, not over interleavings

Proposition 17.1 (Causality). *The value of a term must be summed over causal histories, not over interleavings. Independent redexes must have their concurrency square filled.*

Failing to do this inflates the value of n independent sites by a factor $n!$, counting the linearisations of a partial order as if they were distinct histories. This is not hygiene; it is load-bearing. Measured: for $n = 2, 3, 4$ independent gadgets, the interleaving inflation is exactly 2, 6, 24, while the causal computation returns squared norm 1.000000 [2].

17.2 The join is the recombiner

The first design question in a graded setting is how to get *constructive* interference at all — how two derivations can recombine so that their values add rather than staying in separate branches. In an ordinary race they cannot: the losing message survives, and that surviving message is a record of which branch was taken.

The answer needs no new term former.

Theorem 17.2 (Scattering). *An m -fold join $\text{for}(y_1 \leftarrow x \ \& \ \cdots \ \& \ y_m \leftarrow x)$ on a channel carrying m data has $m!$ candidate matches — the bijections from patterns to data — all of which consume all the data. There is no residual message and no which-path record. Writing A_{ij} for the clause’s value on assigning datum j to pattern i , the symmetric outcome carries $\text{Per}(A)$.*

Corollary 17.3 (Which-path erasure is decided by the continuation). *The candidates’ outcomes coincide exactly when the continuation is symmetric in the bound names. No clause produces interference if the continuation distinguishes its bound names. Whether a race can interfere is therefore a property of the continuation, not of the weights.*

Symmetry is cheap to arrange, because parallel composition is commutative: $\mathbf{z}!(\ast \mathbf{y}1) \mid \mathbf{z}!(\ast \mathbf{y}2)$ is symmetric and $\mathbf{z}1!(\ast \mathbf{y}1) \mid \mathbf{z}2!(\ast \mathbf{y}2)$ is not. The minimal interferometer is three lines of rholang, and it is the Hong–Ou–Mandel experiment: one detector channel versus two. Taking the clause to carry the sign of the permutation gives $\text{Det}(A)$ instead, and fermionic statistics.

Principle 17.4 (Interference is a false disjunction with true disjuncts). *In the crisp setting a disjunction with a true disjunct is true. Once $\oplus$ has additive inverses this fails, and the failure is exactly cancellation. Interference is not an addition to the logic; it is what $\oplus$ does when the value algebra permits it.*

17.3 The cost: the language is too powerful, not too weak

Grading over $\mathbb{C}$ makes the guard language expressive enough to write quantum circuits — a one-qubit gate is a join over an input channel and a two-datum ancilla channel, with the clause supplying the four matrix entries; a two-qubit gate is the same construction with four candidates and an arbitrary 4×4 ; deterministic communication has a single candidate of weight one and is therefore amplitude-transparent, so ordinary rholang runs inside every branch and the classical part of a quantum algorithm costs nothing special [2].

The problem is the other direction.

Proposition 17.5 (Overshoot). *The semantics is unnormalised with the Born rule applied at observation. Non-isometric clauses therefore change the norm, and renormalising at the end is post-selection. Unrestricted graded rho with complex clauses is thus $\text{PostBQP} = \text{PP}$ [44] — strictly more than a quantum computer.*

The response is a discipline rather than a retreat: declare a coherent mode in which clauses must be *isometric*, and check the condition statically. Isometry is decidable by inspection for clauses in the normal form of Theorem 16.1 with constant scalars, and not obviously so for clauses using the graded modality. That stratification — *the modality-free fragment with isometric clauses is the extractable fragment* — is what a circuit compiler can consume, while the modality remains available in the stochastic and probabilistic regimes where no unitarity is expected. Part V shows that this same fragment is characterised in a completely different way, as the fragment that demands no choice.

18 Instances

$\mathbb{R}_{\geq 0}$: propensity. The value is a rate; the total propensity of a configuration is the sum over funded redexes; the resulting dynamics is a continuous-time Markov chain and the simulation is Gillespie’s. Redex multiplicity must be counted — redexes are selections from a multiset, not terms up to structural congruence — or the simulator runs at the wrong rate; the mistake is invisible in a trace and was caught only statistically [12].

$[0, 1]$ with a probabilistic logic. Values supplied by a probabilistic logic network are pairs (strength, confidence). Revision in such a logic is *not* a semiring addition — it is idempotent, unit-free and non-distributive — so the honest arrangement is that the logic supplies values and $\mathbb{R}_{\geq 0}$ supplies the algebra, with the clause taking the product $s \cdot c$. This is exactly multiplicative under deduction in the high-uncertainty regime [43]. Confidence is not discarded: it enters the propensity, so weak evidence yields a rarely-taken branch rather than a forbidden one.

A quantale. The quantale-graded calculus [14] is the setting where the graded sort loses nothing: complete lattice, hence Knaster–Tarski, hence graded fixed points with no convergence side-condition, and — if the quantale is Girard — negation as well. Over $\mathbb{R}_{\geq 0}$ the fixed point becomes a convergence question with a spectral-radius condition; over $\mathbb{C}$ there is no order and the fixed point becomes a budget.

$\mathbb{C}$. Amplitudes, interference, and the discipline of §17.

19 Negation, assembled

Negation has now appeared as an obstruction five times, in five apparently unrelated places. Collected, they are one phenomenon.

1. **Adequacy needs it.** A generated logic separates exactly what interaction separates only if the target specification supplies something like both a conjunction and a negation. Generators that emit type systems supply neither, and offer no adequacy theorem (§3).

2. **The assay needs it.** The complementary guard pair literally writes $\neg\varphi$ in the second receipt, so the mortal scientist's central experimental apparatus depends on the hypothesis logic having a negation (§14) — a dependency the original presentation did not advertise.
3. **The graded sort cannot have it in general.** $1 - x$ works on $[0, 1]$; $\mathbb{C}$ offers no canonical candidate and De Morgan fails (§15).
4. **The container says why.** A quantale supports involutive negation exactly when it has a dualising element, i.e. when it is a Girard quantale. $[0, 1]$ with Łukasiewicz negation is one; $\mathbb{C}$ is not even a quantale (Observation 9.7). Items 2 and 3 are the same fact seen from the two ends of the value dial.
5. **Choice classicalises, and interference is the casualty.** By Diaconescu's argument, full choice implies excluded middle; a value algebra with excluded middle reads its clauses crisply; and a crisp reading has no additive inverses, hence no cancellation. A resolver strong enough to realise full choice can exhibit no interference deficit [39, 3]. The converse would need a topos of graded predicates and is open.

Principle 19.1 (Negation is the connective that trades adequacy against gradability). *Supplying negation buys adequacy and the complementary guard pair, and costs cancellation. Withholding it buys interference and costs the ability to state a refutation directly. The trade is not avoidable by cleverness; it is a property of the value algebra, and the criterion for which side one is on is whether that algebra is a Girard quantale.*

There is one graceful escape, and it is the reason the probabilistic instance matters more than as an application. In a Girard quantale of the Łukasiewicz kind, refutation is $1 - s$, which always exists; and confidence supplies a second coordinate in which the undetermined outcome lives. The assay's trichotomy of confirm, refute and undetermined therefore survives as a theorem, but as three *corners of a square* rather than three cases of a case-split — and the undetermined outcome stops being honest residue and becomes a coordinate. That is the change that lets the forager of Part VI learn.

Part V — Price

A graded clause assigns values to the candidate resolutions of a race. It does not pick one. Something else has to, and this part is about what that costs. The distinction between weighing and selecting turns out to be the quantifier/selection distinction of Escardó and Oliva; the condition under which a selection is definable is a condition on the value algebra; and the strength of the choice principle needed to resolve a given clause can be written down, one formula per degree of the Weihrauch lattice, in the same syntactic slot. The part closes with a conjecture identifying this grading with the packaging grading of Part II, a failure of subject reduction, and its repair. Source: [3] throughout, with [4] for the programme it answers and [40] for the categorical backbone.

20 Weighing is not selecting

Write $\text{Cand}(P, r)$ for the set of candidate matches of rule r at term P . A graded clause is a map

$$\psi : \text{Cand}(P, r) \longrightarrow V,$$

a *weighting* of the fibre. A scheduler is a map

$$\varepsilon : \text{Cand}(P, r) \longrightarrow \text{Cand}(P, r) \text{ (a chosen element),}$$

a *section*. These are different animals, and conflating them is the error the whole of this part is organised to avoid.

The categorical statement is due to Escardó and Oliva [40]. A clause is a *quantifier*: an element of the K -monad $(X \rightarrow R) \rightarrow R$. A scheduler is a *selection function*: an element of the J -monad $(X \rightarrow R) \rightarrow X$. There is a canonical morphism $J \rightarrow K$ sending $\varepsilon \mapsto \lambda p. p(\varepsilon(p))$, and the statement that a particular resolver realises a particular clause is the statement that this morphism sends one to the other.

Principle 20.1 (Division of labour). *The clause is the family and its measure. The choice principle is the assertion that a section exists uniformly. The scheduler is the section. A clause therefore cannot be a choice principle; formulae certify an altitude rather than asserting it.*

This is the reading the programme note asked for. The choice-scheduling development [4] proposes a Curry–Howard correspondence in which choice principles are types and fair schedulers are terms, and records that the choice type of a cut is a semantic invariant, not in general computable, and “approximable from above by types”. The graded clause is that effective approximation.

21 Attainment

When is a section definable, so that the resolver can actually hand back a candidate rather than an expectation?

Proposition 21.1 (Attainability). *(i) If $\oplus$ is idempotent and ψ is crisp, a definable section exists.*

- (ii) If $\oplus$ is idempotent and ψ is graded, the value is attained exactly when the winning candidate carries the multiplicative unit.
- (iii) If $\oplus$ is not idempotent, the clause is an expectation quantifier and is never attained into Cand.

Measured over 20,000 payoff draws per algebra: $\mathbb{B}$, Viterbi and tropical all attain 100% of the time on crisp clauses; Viterbi attains 73.83% on graded clauses; tropical attains 0%; $\mathbb{R}_{\geq 0}$ attains 0%. Clause (ii) was confirmed at 100% agreement [3].

Remark 21.2 (Restriction versus discount). Gating is free; preference costs. A clause that merely says which candidates are eligible leaves attainment intact; a clause that says which are *better* generally does not.

Non-attainment is not a defect. It is a *lift*: over $\mathbb{R}_{\geq 0}$ the resolution lifts to distributions and is attained in expectation, verified by Monte Carlo at 0.590826 against an analytic 0.590000 and 0.390489 against 0.390000 over 400,000 draws.

Principle 21.3 (The lift is the oracle). *Where a section is not definable, the resolver must be handed something extra — a source of randomness, an oracle, an aperture. The type of that extra thing is what the Weihrauch degree measures.*

Attainment is also where Part II’s coalition row was decided. Tropical and Viterbi are exactly the two algebras where clause (i) holds on the nose; they are also the two picked out by the quorum semantics; and they are therefore the algebras in which an observer of a population receives a witness coalition rather than a scalar (Observation 10.4).

22 The graded fixed point, and three engines of ascent

The graded sort as introduced in Part IV is finitary, and a finitary clause cannot climb. Adding fixed points is what makes altitude available:

$$\psi ::= \cdots \mid X \mid \mu X.\psi \mid \nu X.\psi \mid \langle @ \rangle_V \psi$$

with the usual guardedness condition. Read μ as *search* and ν as *survival*.

Proposition 22.1 (Existence by algebra). *A quantale gives the fixed point by Knaster–Tarski. Over $\mathbb{R}_{\geq 0}$ existence becomes a convergence condition (spectral radius below one). Over $\mathbb{C}$ there is no order and the condition is spectral or, in practice, a budget.*

Budget is the honest default, and it is a semantic dial: spend budget, gain altitude.

Proposition 22.2 (Three engines). *There are exactly three ways for a clause to climb: depth (nesting a modality under ν), width (quantifying over redex positions under μ), and reflection (the modality $\langle @ \rangle$ over quoted processes). Depth and width are Weihrauch-incomparable — they correspond to $\mathbb{C}_{\mathbb{N}}$ and $\mathbb{C}_{2^{\mathbb{N}}} = \text{WKL}$ — and that incomparability is the syntactic source of the order being a lattice rather than a chain.*

That last clause is worth pausing on, because it converts an assertion into a derivation. Both [4] and the agency note say the order of choice principles is “a jumble, not a tower”. Proposition 22.2 says *why*: there are two independent syntactic engines, and neither dominates the other.

23 One formula per degree

The formulae below all live in the same syntactic slot — a **where** clause — and differ only in shape. This is the concrete sense in which one programming paradigm covers the whole hierarchy.

Three entries deserve comment.

LLPO by accident. A race between two messages whose guards are semi-decidable is already at C_2 . This is not an exotic construction; it is what an ordinary concurrent program looks like. Verified computationally: over $3 \times 40 \times 60$ instance/stage pairs there were zero empty supports, and every stage- k selector, for $k = 0..7$ and both tie-breaks, was defeated by the diagonal that produces at $k+1$ on the committed side.

Dependent choice is free. Proposition 16.3 said that a clause evaluated against the current term makes later values depend on earlier firings with no extra mechanism. In the present vocabulary that sentence reads: later fibres depend on earlier selections. It is the principle of dependent choice, and it required no new syntax at all.

The bottom of the lattice is the extractable fragment.

Proposition 23.1 (Bottom). *Every fixed-point-free clause has choice type $\perp$.*

Corollary 23.2 (Two characterisations of one fragment). *The fragment a circuit or diagrammatic compiler can consume — modality-free with isometric clauses (§17) — and the fragment that demands no choice coincide. The compiler consumes what it consumes because that fragment demands no choice.*

24 Reaching the top, and the two conspicuous routes

Proposition 24.1 (Ceiling). *For a closed term over presented channels with no reflective modality under a fixed point, the choice type is at most $C_{\mathbb{N}}$.*

The reason is that names are quoted processes, hence countable, and the store's bags are finite and presented; a where-clause over a presented store simply cannot reach full choice. Two routes escape.

Route E: the aperture. A channel fed by the environment rather than by the term is not presented. This is the syntactic form of the observation that capacity lives at the apertures, and it doubles as a security statement: a static analysis of aperture channels bounds the choice strength of an entire shard.

Route R: reflection. The clause $\nu X.(\langle @ \rangle_V X \otimes [\beta(*y)])$ is impredicative, because the domain of the reflective modality contains processes that may carry the clause itself.

Corollary 24.2 (Trichotomy). *There are exactly these two routes above the ceiling, and both are syntactically conspicuous.*

Remark 24.3 (Krivine’s quote). Krivine realises dependent choice by adding a **quote** instruction to the machine. The **rho** calculus *has* quote as a term former, and has had it since 2005 [15]. Reflection is the calculus’s native choice primitive. This is the single most compact statement of why the calculus is the right host for this analysis, and it is also a warning: the same primitive is the suspect in the failure of subject reduction below.

25 Interference is transverse

Proposition 25.1 (Transversality). *A non-empty interference deficit is not the image of any section, because a section always lands in the fibre. The complex case is therefore off the Weihrauch lattice, not high on it.*

Recomputed: phases 0, π and $\pi/2$ give squared coefficients 2.000000, 0.000000 and 1.000000, while over $\mathbb{R}_{\geq 0}$ the minimum over 100×100 strictly positive pairs is 0.020000 and never zero.

Combined with item 5 of §19, this gives a design rule rather than a paradox: an aperture may host a resolver of high choice strength, and the interior may host a coherent resolver, but no single resolver can do both jobs.

26 Two gradings, one measurement?

Two Weihrauch gradings have now appeared in this document, derived independently and for different reasons.

The first is in Part II, from [1]: the degree of PACK_S , the operation packaging a container’s splittings back into an object of the same shape. Its choice residue for **Bag** is a linear order on atoms; for P_{fin} it adds decidable equality; for **List** it is the identity, because term positions are already ordered.

The second is this part’s: the degree $\chi(c)$ of the choice principle needed to resolve a race under a given clause. Its ceiling for presented channels is $\mathbf{C}_{\mathbb{N}\mathbb{N}}$, because names are quoted processes.

Conjecture 26.1 (The two gradings agree). *They are the same measurement. The candidate set of a where-clause is a bag — the store’s produces and consumes — and selecting a match is packaging that bag into a distinguished element, i.e. pack_{Bag} . Accordingly the choice type of a clause is the packaging degree of its candidate container, and Proposition 24.1 is Principle (trade) of [1]: a presented store bounds the resolution because term positions are already ordered.*

We state this as a conjecture and do not lean on it. The evidence is that the two bounds have the same content in different words, that both turn on presentedness, and that both locate their escape in the same place — a domain that is not presented, which on one side is an aperture and on the other is an atom object too large to order syntactically. The obligation is to make “the candidate container” precise for clauses using the graded modality, where the candidates at one site depend on transitions available at another.

Remark 26.2 (A related conjecture, better supported). A second identification, verified computationally though not proved, is that the Escardó–Oliva product of selection functions, Weihrauch composition, and chaining graded sites through the graded modality are one operation. Over 20,000 tables there were zero disagreements, with a negative control — chaining in the wrong algebra — producing 4,483. The content of the agreement is that clause algebra and scheduler discipline must match.

27 Subject reduction fails, cheaply, and how to repair it

Proposition 27.1 (Strength can be mailed in). *Let $P = c!(Q) \mid \text{for}(y \leftarrow c \text{ where } [\text{true}]) *y$. Every clause in P is fixed-point-free, hence at $\perp$ by Proposition 23.1; but P reduces to Q , which may be at WKL. Choice type is not preserved by reduction.*

The mechanism is reflection: substitution replaces y by a quoted process and $*y$ drops an arbitrary received process into term position, so a low-altitude clause can receive a high-altitude one.

Definition 27.2 (Stratified **where**). Declare a ceiling L and require a level check on drops: a received process may be dropped only if its clauses are at or below L .

Proposition 27.3 (Subject reduction, stratified). *Under Definition 27.2, choice type is preserved.*

Corollary 27.4. *In a stratified shard, full choice occurs exactly at declared apertures, and the interior's choice type is invariant.*

Remark 27.5 (The same suspect, four times). Reflection is now behind: the possibility of length-asymmetric reconvergence and hence of a second, non-local source of interference; Route R to full choice; the failure of subject reduction above; and the corresponding question for the graded coalition modality, where a received process can mail in a high-altitude coalition formula. Krivine's **quote** is the same primitive doing the same two jobs — supplying choice strength and breaking invariance — and it is not a coincidence that a calculus powerful enough to be its own metalanguage has to pay for it here.

28 Fairness is a formula

One last consequence, which moves a familiar concept from the typing side of the correspondence to the term side.

Proposition 28.1 (Zero is absorbing). *A continuously enabled receipt whose clause evaluates to zero never fires, and therefore never acquires the evidence that would make its clause non-zero.*

This is not hypothetical. In the foraging study, a learner beginning with no prior evidence has confidence zero, hence clause value zero, hence a receipt that is permanently enabled and never fires: zero openings, terminal wealth 520.0, which is the do-nothing baseline, against 3543.7 for the same learner given a single unit of prior evidence [2]. Not death, but stasis — manufactured by the value algebra.

Definition 28.2 (Exploration combinator). $\psi^\varepsilon := \psi \oplus \varepsilon$ for a constant $\varepsilon > 0$: *exploration is a disjunct.*

Proposition 28.3 (Weak fairness is a clause combinator). ψ^ε *enforces weak fairness at the site.*

Measured: with no disjunct, 0 firings and terminal 0.00; with $\psi \oplus 0.02$, 179 firings and terminal 83.80; with one unit of prior evidence instead, 197 and 89.20.

Remark 28.4 (Strong fairness has a degree, and the degree is compactness). Strong fairness is a ν over the run tree, hence the depth engine, hence **WKL**. The classical weak/strong fairness gap thereby acquires a Weihrauch degree, and the content of the gap is compactness. Over $\mathbb{C}$ the ε -disjunct can cancel, so the fairness repair is a de-tuning device in the coherent regime and not a guarantee.

Slogan 28.5. *A shard can have any two of: unrestricted reflection, invariant choice types, and coherent resolution.*

Part VI — The range, worked

Six observation disciplines, each a different setting of the dials of Parts I and IV, each with an example that has been run. The purpose of the part is to make the word “range” concrete: these are not variations on a theme but logics that see different things, cost different amounts, and support different kinds of learner. Numbers are quoted from the notes and their simulations [8, 9, 2, 5, 11]; nothing has been re-run here.

29 How to read this part

Three conventions recur, and it is easier to state them once.

Pricing. An observation is charged by the depth of the formula that performs it, on the schedule $\kappa(d) = 2^d$ per candidate examined. The schedule is a choice, and it was varied: the qualitative results below are robust to it and the exact boundaries are not. Reading structure the environment has already deposited is charged at the formula’s spatial depth; driving the system forward to find something out is charged per rendezvous, and costs more.

Currency. Where a learner is embedded in an ecology, results are reported as *yield per metered unit* — how much better the learner does per unit of observation it buys. This makes claims about theories comparable to claims about food, which is the point.

Ladders. A *ladder* is a sequence of strategies, each identified with a formula, ordered by the depth and modal complexity of that formula, with the exactly-correct characteristic formula χ at the top. A ladder makes the observation discipline into an independent variable.

30 Discipline 1 — Spatial, crisp

Dials: $S = \mathbf{P}_{\text{fin}}$, $V = \mathbb{B}$, *structural connectives only, no modality.* This is perception: predicates about the shape of the term, with no lookahead.

The worked example is noughts-and-crosses [8], over the complete census of 255,168 plays — 5478 reachable positions, 765 up to symmetry, 4520 non-terminal, of which 1890 (41.8%) offer a fork.

The instructive result is not about play but about *encoding*. A fork is two threats that cannot both be blocked, and expressing it requires graded separation at $k = 0$: $\text{Threat} \mid_0 \text{Threat}$, two threats sharing no gap. Requiring that formula to be expressible *forces the board encoding*: the grading namespace must be indexed by squares, not by lines, because two threats with a common gap must be excluded and only a square-indexed namespace can see the sharing.

Observation 30.1 (The logic constrains the model, not only the other way round). *Which predicates one wants to be able to state is a constraint on how the world is represented. This is the first and simplest sense in which choosing an observation discipline is a commitment rather than a convenience.*

The purely spatial rung is also the best purchase on the entire ladder: at modal depth 0 and name-level depth 0 it returns +0.0082 win per metered unit, roughly five hundred times the rung below it.

31 Discipline 2 — Modal, crisp

Dials: as above, plus the context-labelled modality. This is interaction: looking ahead by driving the system.

Getting this right required the correction mentioned in §2. The context K must be a genuine redex context, which forces an empty square to be a receipt offering to be claimed and an occupied one a resting send, so that a ply is exactly one rewrite and $\langle K_s \rangle$ is a generated primitive rather than a derived abbreviation. Redex positions then correspond bijectively to empty squares, and the branching factor of $\exists s$ is the legal move count.

Four results, all exact over the full census.

Truth is dominated. χ wins .873 at cost 679.2; rung 5 wins .918 at 255.9. χ is the unique rung that never loses — truth buys safety, not food — and it invades a rung-5 population only if the endowment exceeds 12,474 times the unit price. Across the whole phase grid it is optimal in two cells.

Perception beats interaction, quantitatively. The three non-modal rungs (1, 2, 5) deliver +0.444 of the total +0.481 gain — 92.3% of the yield — for 87.9 of 255.9 units, 34.3% of the cost. The two modal rungs deliver the remaining +0.038 for 168.0 units. Rung 4, the one genuinely alternating rung, is measurably the worst purchase on the ladder.

Principle 31.1 (Read or drive). *There are two routes to any predicate: read it, if the environment has already computed it into legible structure, priced by formula depth; or drive it, priced by rendezvous. This is a claim about environments, not about learners, and it is statable only because the framework prices the grades separately and can therefore ask who is billed.*

Confinement, measured. The best opening reverses with theory depth: rung 5 prefers the centre (.833 against the corner's .597), χ prefers the corner (.896 against the centre's .548). Deepening the theory while holding the opening fixed takes the learner from .833 to .548. The corner-opening mutant invades at identical cost and is exactly the distance-2⁻¹ move that Confinement denies an individual.

A solved science is a dead ecology. A monoculture of χ , and a monoculture of centre-opening rung 5, both draw every game, so the harvest rate is 0.000 and the population starves on a finite wall. The corner-opening rung-5 population sustains .333 and lives. Imperfection is load-bearing.

And the inversion. Run the same construction on Nim [9] and the headline reverses. In Nim(1, 2, 3) the rungs give win rates .5000/.5917/.7958/.9000 at costs 0/15.2/31.4/38.5, with returns *rising*: +.00602, +.01262, +.01476. Here χ — move to nim-sum zero, a name-level depth-0 formula — is the cheapest rung on the ladder and the best buy on it. In Nim(3, 4, 5) the returns are flat.

Observation 31.2 (Affordability is a property of the world). *Whether truth is worth its price is not a fact about the learner. Nothing about the learner changed between the two games. What changed was the environment’s structural regularity, and with it whether the correct theory is cheap to state.*

32 Discipline 3 — Graded, real-valued

Dials: $V = [0, 1]$ with probabilistic-logic values, clause value the product $s \cdot c$ of strength and confidence. This is the discipline in which a learner can be disappointed.

The world [2]: specimens are rich ($\sigma_R = 120$) or decoys ($\sigma_D = 0$); opening costs $\kappa_{\text{break}} = 40$; a depth-2 structural test χ costs $\kappa_{\text{sense}} = 4$; grazing yields 5; endowment 120; horizon 400. World A has $P(\text{rich}) = .40$, $P(\chi \mid R) = .8$, $P(\chi \mid D) = .3$, so the posterior is .64 and $E[\text{net} \mid \text{open}] = +36.80$. World B has $P(\text{rich}) = .15$ and $P(\chi \mid D) = .4$, posterior .26, $E = -8.70$: opening on the test is a losing proposition.

The bootstrap trap. A learner with no prior evidence has confidence zero, hence clause value zero, hence a permanently enabled receipt that never fires and therefore never acquires evidence: 0.00 openings and terminal wealth 520.0, the grazing baseline, against 3543.7 with a single unit of prior evidence. This is Proposition 28.1 in the wild, and its remedy is Definition 28.2: exploration is a disjunct.

Confidence buys variance, not yield. In world A at scepticism parameter $k = 4$: the crisp forager has survival .8501 and terminal 4084.7 — the highest terminal wealth measured and the lowest survival — against the graded forager’s .9485 and 3543.7.

And in a harsh world it is not variance but existence. In world B the crisp forager’s survival is .0029. It cannot learn to stop, because there is nowhere for the disappointment to go: the formula χ was true and remains true. The graded forager’s policy is the formula’s *value*, and the value moves — strength $.750 \rightarrow .250$, confidence $.333 \rightarrow .600$, product $.250 \rightarrow .150$, probability of opening $.333 \rightarrow .231$ over five openings.

Amortisation, computed. Against the true posterior .6400: a learner whose hypotheses are about individuals, so that evidence cannot accumulate past $n = 2$, has mean error .2039; a learner whose hypotheses are about classes reaches $n = 101.54$ and mean error .0383 — better by a factor of 5.3, for evidence that costs nothing extra. This is the defensible version of a claim the original note overstated: class hypotheses have an amortisation advantage, not a monopoly. Individual hypotheses are not valueless; they outlive the crisp forager in both worlds.

Scepticism has a metabolic optimum. Under log terminal wealth — the right criterion for a multiplicative absorbing process — world A peaks at $k^* = 9$ (7.8741, against 7.744 at $k = 4$ and 7.715 at $k = 20$), and a 50/50 ensemble of the two worlds peaks at $k^* = 64$ (6.4309, against 6.343 at 32 and 6.263 at 512). World B alone runs to the boundary, because abstinence is right there.

Principle 32.1 (Scepticism is priced by the environment). *A parameter that in the source logic is set by taste — how much evidence it takes to become confident — acquires a selection criterion*

once the learner is mortal, and the optimum rises roughly sevenfold with the riskiness of the world.

Remark 32.2 (Confidence needs an outside option to be visible). If the competing branch is the complement of the same belief, confidence cancels exactly: $sc/(sc + (1 - s)c) = s$. Grading a guard buys nothing unless the learner has something else to do. In the forager that something else is grazing.

33 Discipline 4 — Graded, complex-valued

Dials: $V = \mathbb{C}$, *clauses isometric, modality-free*. This is the coherent regime, and its example is Shor’s algorithm written in the guard language [2].

The construction is the join gadget of Theorem 17.2: a qubit is which datum sits on a channel; a gate is a join over the input channel and a two-datum ancilla channel with both ancilla patterns joined, so everything is consumed and the outcomes coincide across input states; the clause supplies the matrix entries. Two-qubit gates take four candidates and an arbitrary 4×4 , and entangle. Gadgets compose in series on fresh channels, and the term’s causal order *is* the circuit’s directed graph — a layer-2 gadget is not enabled until its input channel carries a datum, so data dependency enforces circuit order with no scheduler.

Measured: the quantum Fourier transform assembled from gadgets matches the DFT matrix to worst-entry error 2.7×10^{-16} , 1.4×10^{-15} , 3.8×10^{-15} at $n = 2, 3, 4$. Modular exponentiation preserves norm to 1.000000000000 on a 16-term superposition — deterministic communication has one candidate of weight one, so the classical part of a quantum algorithm costs nothing special in this semantics. For $N = 15$, $a = 7$ with $4 + 4$ qubits: eight one-qubit and eight two-qubit gadget firings, 390 candidate evaluations, 16 non-zero configurations, and the counting register at 0/4/8/12 with probability exactly 0.250000000 each, totalling 1.000000000000; continued fractions give $r = 4$, and $\gcd(7^2 - 1, 15) = 3$, $\gcd(7^2 + 1, 15) = 5$.

Observation 33.1 (Shor is an interference deficit arranged on purpose). *Twelve of the sixteen reachable outcomes carry amplitude zero. Corollary 15.2’s inclusion $\text{supp} \subseteq \text{Reach}$ is strict by exactly those twelve. What a quantum algorithm is, in this semantics, is a large deliberate strictness.*

The honest limits are worth repeating. The encoding is circuit-shaped: every gate is a hand-built join gadget, and mobility, reflection and name-passing all idle. It shows expressiveness, not a native idiom. Writing Shor is not running Shor; classical evaluation of the denotation is exponential. And by Proposition 17.5 the unrestricted language is PP, so the isometry discipline is what makes this a quantum language rather than a magic one — measured on a post-selection filter $\text{diag}(1, 10^{-3})$ after a Hadamard layer, where the surviving norm falls as $2^{-n/2}$ while the renormalised probability of the all-zero outcome climbs to .999996, .999994, .999992, .999990 for $n = 4, 6, 8, 10$.

34 Discipline 5 — Semitopological

Dials: $S = \text{Open}(P)$, V tropical, *the coalition modality of §10*. This is the discipline of populations, and it is worked twice: once with the observer outside the population, once inside [5].

34.1 Outside: a pride reading a herd

Model: a ring of $N = 24$ animals, neighbourhood window 2, noise .03, asynchronous majority updating. Opens are unions of contiguous arcs of length at least k — a genuine semitopology, since two length- k arcs can meet in a set that is not open. Pricing $\kappa(d) = 2^d$: a heading costs 2, deposited confidence 4, testing an open +4, one driven tick 8.

Seven rungs, over 4000 trials at $t = 90$, $k = 13$, reported as coverage / precision / accuracy / cost: R_0 1.0000/.4985/.4985/0; R_1 1.0000/.9002/.9002/2; R_5 .8498/.9744/.9031/4; R_2 1.0000/.9868/.9868/10; R_3 .7510/1.0000/.8755/190.2 (return -0.00062); R_4 identical at 192; R_6 1.0000/.9832/.9832/1344.

Observation 34.1 (The modality buys irreversibility, not accuracy). *Across 6839 coalition verdicts at five different times, precision was 1.0000 at every one — including at $t = 30$, where coverage is 0.37%. Sample majority over the same range runs .9160 to .9873. The expensive modal rung does not predict better; it predicts only when it is sure, and declines otherwise.*

Principle 34.2 (A second currency). *Yield per metered unit underprices an instrument that declines to answer. An observer facing an irreversible commitment — a charge, a takeoff — wants precision conditional on answering, not accuracy overall, and will pay a great deal for it.*

The counter-move, and what it tells us about privacy. Insert seams — zero-coupling bonds — into the ring, from 0 to 6. The herd's own probability of reaching a conclusion falls from .7257 to .2331; the pride's coverage falls from .7520 to .0115; the pride's precision stays between .9986 and 1.0000. Illegibility is a denial of *liveness*, not an induction of error — exactly the signature of a consensus failure seen from outside.

Principle 34.3 (The legibility trade). *Safety versus liveness, and cohesion versus illegibility, are one trade-off. A population buys privacy with the currency it buys agreement.*

Compression, and the clock. A topen — a set of participants guaranteed to decide alike — is maximal compression for an observer: inside one, prediction accuracy is .8683 over 6.5 predictions per observation; across a seam it is .5606, which is chance. And the graded reading answers a question the crisp one cannot ask: bucketing by margin quartile, the ticks-to-commitment run 42.4/29.6/28.9/18.0 with direction correct at .6080/1.0000/1.0000/1.0000. *The margin is a clock.*

A negative result, kept. At $k = 13$ the tropical $\otimes = \min$ over a coalition is zeroed by a single dissenter, so the crisp and graded rungs return identical rows. Grading pays only if the coalition size is relaxed as well: two dials, not one. The result is kept in the note as a labelled negative.

34.2 Inside: a swarm that decides

Honeybee nest-site selection [41, 42] is the resolute case: the scout is a participant evaluating a coalition formula about its own population, and the detection *is* the resolver.

Proposition 34.4 (The mechanism is forced by altitude). *Consensus-at-the-cluster is a global universal with no presented witness — finitely refutable but not finitely confirmable. Quorum-at-a-site is a local $\oplus \otimes$ over one open. No bee can evaluate a global universal, so the swarm must use the local formula.*

That is Seeley and Visscher’s empirical finding [42] derived from a cost argument rather than assumed. The swarm also exhibits the split of §20 as different behaviours: dance evaluation is the clause, piping and takeoff are the selection function, and no bee does both.

Theorem 34.5 (The $2k$ theorem). *Two simultaneous quorums require $2k$ committed scouts. A swarm with active pool $S < 2k$ cannot split, for any site qualities, recruitment rates, or level of cross-inhibition.*

Measured, with $k = 15$: split frequency is identically 0.0000 at $S = 12, 16, 20, 24, 28, 29$, first appears at $S = 30 = 2k$ at 0.0130, and rises to 0.6850 at $S = 50$. Takeoff-with-dissent, observed occasionally by Lindauer and by Seeley, is thereby confined to $S \geq 2k$.

Principle 34.6 (Cross-inhibition changes jobs at $S = 2k$). *Below the threshold it prevents deadlock; above it, it prevents splitting. Measured: at $S = 16$, deadlock falls from .5120 to .0000 when inhibition is switched on, with no splits in either column; at $S = 40$ there is no deadlock in either column and splitting falls from .3860 to .1170. The literature credits stop signals with both roles and does not separate them; this is the discriminating experiment.*

Two bottoms, and a threshold that is not a number. A scout cannot distinguish “quorum not reached” from “I have not sampled enough”. Under encounter sensing, false positives are impossible and all error is false negative, and the apparent threshold scales with effective cavity area: measured 16, 21, 25, 33 for areas 1, 1.25, 1.5, 2, giving ratios 16.0/16.8/16.7/16.5 against a nominal $k = 15$.

An unexpected cost, recorded honestly. With cross-inhibition on, the swarm picks the better site slightly *less* often — .7235 against .7845 at $q_B = 0.8$ — because inhibition suppresses correct advocacy too. The benefit of inhibition falls from .3140 to .1950 as the site disparity widens. This was not predicted and is flagged as open.

Remark 34.7 (Position statement). None of this out-predicts the evidence-accumulator models on trajectories, and it is not a competing dynamical model. Its contributions are a logic for what the bee is testing, a cost argument for why it tests that rather than something else, and structural impossibility results — Theorem 34.5 — that a statistical model cannot state. The swarm predictions are untested proposals.

35 Discipline 6 — Resolution-graded

Dials: any of the above, with the observation depth deliberately capped. The question is no longer what a discipline can see but what an observer *using* it reports about a world it cannot fully resolve [11].

Counting requires a scope, and a scope with no bound is not a computable object. The proposal is that a *namespace* supplies the computationally coherent notion: a scope is a predicate, not a set; disjointness of scopes is structure rather than a limitation, since $@(\varphi \mid \psi)$ with φ and ψ disjoint parses uniquely and description length is additive; and fractality is recovered by a fixed point on namespaces, $\mu X.@((\varphi \vee X) \mid (\psi \vee X))$. Membership in a μ -generated scope is decidable by descent — disjointness kills the search and the no-self-code theorem makes the descent well-founded — so the computational coherence of scope comes from reflection, not from finiteness.

Copy number is then graded by the resolution n at which the observer can separate kinds, and the received integer is the flat collapse of a *copy series*. For a worked individual with four skills — forage ($p = .95$, $d^* = 3$), pod ($p = .99$, $d^* = 5$), mate ($p = .90$, $d^* = 3$), rear ($p = .99$, $d^* = 5$) — comprising 96 learners, in a pod of 12:

The assembly floor for the individual is 28 with skills sharing structure and 43 without — a saving of 34.9% — and 57.1% of the floor is content rather than architecture. The most expensive skill is the least reliable one, not the deepest.

Principle 35.1 (A learner reports a world with fewer kinds in it than the world has). *An observer does not fail to see the distinctions it cannot afford. It fails to see that they are there. The failure is invisible from inside the discipline, and the framework can compute its size.*

Remark 35.2 (How far the claim goes). An earlier and stronger version of this — “it takes a mind to find a mind” — is too strong. The observer must match the depth at which *kinds* separate, which is not the assembly index of the object. The defensible claim is about instruments: a survey needs the second-deepest separation, since at the deepest one the last kind comes free.

36 The measured results, collected

Part VII — Learning, and the fractal

This part cashes the title. A learner holding a graded formula has two degrees of freedom, not one: it can change the formula, or it can change the value. The two live in different spaces, obey different geometries and need different search methods, and the resulting structure is a fibration rather than a space. The part then follows the construction upward — learners compose, and the composite is itself a learner — and closes on what an observation discipline implies about what a mortal thing can know. Sources: [7] for the lattice and the ecology, [9] for composition, [10] for individuation, [11] for the counting.

37 The hypothesis space is a fibration

Recall the two structures now in play.

The formulae of a generated logic form the relaxation lattice, ordered by logical strength and metrised by modal depth: formulae agreeing to depth n are within 2^{-n} . This is an ultrametric, so the balls are nested or disjoint, every point is a centre, and there is no betweenness. There is no gradient (§14). Search is backtracking, priced by how far one had to back up, and Confinement says small steps never leave the initial ball.

The values of a graded clause form the algebra V , which for the learning disciplines of Part VI is $[0, 1]$ or $\mathbb{R}_{\geq 0}$ — ordered, continuous, and navigable by exactly the local methods the ultrametric denies.

Observation 37.1 (Two degrees of freedom). *A graded hypothesis is a pair: a formula and a value. Revising the value is not a move in the ultrametric at all — it is a move of distance zero, since the formula is unchanged. So the hypothesis space is not a metric space with extra decoration; it is a fibration, with the ultrametric tree of formulae as base and a V -valued fibre over each point.*

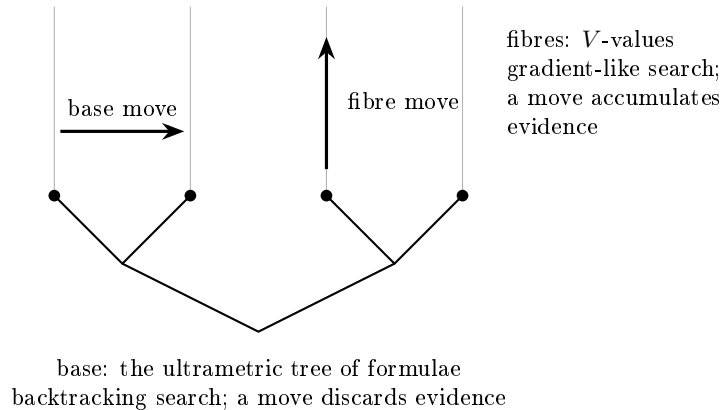


Figure 3: The hypothesis space of a graded learner. Confinement is a theorem about the base; a fibre move has base-distance zero, so it cannot leave a ball — but it can move anywhere inside one.

This is the correct statement of the claim that graded where-clauses are a learning instrument, and it is more modest and more useful than the version one is tempted to assert.

What grading does not do. It does not escape Confinement. Confinement is a theorem about the base, and a fibre move has base-distance zero, so a learner that only ever revises values is confined *absolutely* — it never leaves the ball it started in, because it never leaves the point it started at. The forager of §32 is a case in point: it learns to stop opening, which saves its life, but it never acquires a better test.

What grading does do. It makes the interior of each ball navigable. Before grading, the only way to change behaviour was to change the formula — a discrete jump, priced by backtracking depth, which invalidates every confirmation obtained below that depth. After grading, a learner can change its behaviour continuously while holding its formula, and its evidence, fixed. In the harsh world of §32 that difference is the difference between .0029 and survival.

Principle 37.2 (Two searches, one learner). *The base requires backtracking search; the fibre admits gradient-like search. A learner needs both, and the framework prices them differently: a base move discards evidence, and a fibre move accumulates it.*

There is a pleasing symmetry in the failure modes. A learner that only moves in the fibre is confined to one theory and can only tune it — it is the forager, safe and stuck. A learner that only moves in the base is the ladder-climber of §31, which can acquire deeper theories but pays for each and, as measured, may be made *worse* by depth if it holds its other commitments fixed. Both pathologies are visible in the numbers, and neither is visible without the other coordinate.

Remark 37.3 (Where the two searches meet). The graded fixed point of §22 is the point of contact. μ is search and ν is survival, and both are fibre-side operations whose availability depends on the value algebra — free in a quantale, a convergence condition over $\mathbb{R}_{\geq 0}$, a budget over $\mathbb{C}$. So how far a learner can search *within* a theory is settled by the same choice that settles whether it can state a refutation. Part IV’s negation criterion and Part V’s fixed-point criterion are the same criterion.

38 What escapes the base

If small moves cannot leave a ball, something else must. The corpus identifies three operators that can, and all three are structural rather than epistemic.

Recombination. Reproduction by crossover on quoted code is a base move of arbitrary size. Because the germ is a quotation — inert, transmissible, inspectable — and because the normal form of a process is a tree of communicating pairs, crossover can swap material at homologous loci, producing a formula at a distance no individual revision could reach. The unit of inheritance is a *redex*: a gene is an interaction, not a structure, and linkage is derived rather than posited. And crucially, a revision move and a crossover are *one move* carried along the characteristic-formula map, applied at two timescales and distinguished by accessible distance [7].

Composition. Two learners in parallel constitute a third. This is the subject of §39, and it is the first of the escapes that requires no new syntax at all — parallel composition is already in the calculus.

Death. A population whose members are confined to different balls explores what no member can. This is why the framework insists that a population, not an individual, is the unit of continual learning: not because populations are better, but because Confinement makes a single learner structurally incapable of leaving its basin, and because mortality supplies the retirement criterion that a purely epistemic account has to invent.

39 Learners compose, and the composite is a learner

A learner, on this account, is a distribution over a population of processes together with resource reservoirs; the population constitutes a *namespace*; and learners compose by parallel composition [9].

Theorem 39.1 (Factorisation). *Namespace disjointness, absence of a spanning redex, and factorisation of the composite’s distribution as a product are equivalent. Graded separation at $k = 0$ is probabilistic independence, and the defect of the comparison map is exactly the interaction.*

Overlap is therefore the dial, and it is a vector: one grade per typed namespace — metabolic, source, problem, mating. The familiar ecological relations (neutralism, competition, predation, mutualism, commensalism, theft) fall out as sign conditions on that vector. And one asymmetry is forced:

Proposition 39.2 (Cooperation is epistemic, competition is metabolic). *Overlap confined to the metabolic and source namespaces has non-positive total gain: there is no metabolic mutualism. Positive sum lives only in the problem namespace, via amortised perception — which is Principle 31.1 again, one level up, in the form of the question of who is billed.*

Two further results are worth carrying, because they bear on the design of multi-agent systems rather than on biology.

Non-interference is unstable, and modularity is purchasable. Independently manufactured names can coincide, so disjointness is luck. Rooting each learner’s namespace at its own quotation makes disjointness a theorem, via freshness-by-quotation. The price is a naming discipline.

Competitive dilution is regressive. At a shared rate-limited source, with shares proportional to inverse cost, each learner retains exactly $c_{\text{other}}/(c_{\text{self}} + c_{\text{other}})$ of the deep inquiry it could afford alone — independent of the reservoir, the rate, and the price. In the worked composite, the learner with the incomplete theory loses 66.9% of its affordable depth while the one with the complete theory loses none of its depth and only a third of its assays; at exhaustion the latter holds 5452 against the former’s 1397. Competitive exclusion in epistemic currency, requiring no contact at all.

40 Testimony, and a counterfeiting attack

Once values are graded, learners can exchange *values* and not only observations. In a probabilistic logic, revision adds evidence counts, so confidence is superadditive; and Proposition 39.2 makes metabolism subadditive. The programme’s slogan therefore becomes an inequality rather than a sentiment.

Observation 40.1 (Belief inflation is the epistemic analogue of minting). *Revision assumes independent evidence. Two learners who both heard from a third double-count, and nothing in the framework forbids it. Tokens are conserved; confidence is not. So there is a counterfeiting attack in the problem namespace with no analogue in the metabolic one, and the plagiarism equilibrium becomes a condition on the grading vector rather than a worry.*

This is stated and not developed. It requires carrying strength and confidence as a pair rather than collapsing them to a product, which needs a lax algebra with a canonical aggregation order, and that is a paper of its own.

Remark 40.2 (A belief is a message). In the calculus the belief sits on a channel as data. Testimony and theft of a hypothesis are therefore the same operation as predation, differing only in which namespace the name lives in. That is either an unfortunate consequence or an accurate model of intellectual life.

41 The fractal

The construction repeats. An individual is a learner; a population of learners is a learner; a lineage is a learner over a longer timescale; a composite of populations is a learner again. What makes this more than a rhetorical spiral is that the generating step is identified and priced.

The generating step is the medium. Communication in the calculus is perfect, without loss of generality, because error is modelled by interposition: a communication becomes a send, a channel process, and a receive, where the interposed process may forward, drop, corrupt, read or convert. Those interposed processes form a namespace of their own — and *the channel namespace at one level is the endpoint namespace at the next*. Well-founded downward by the no-self-code theorem, bounded upward by the profitable radius. The outside of one level is the inside of the next [10].

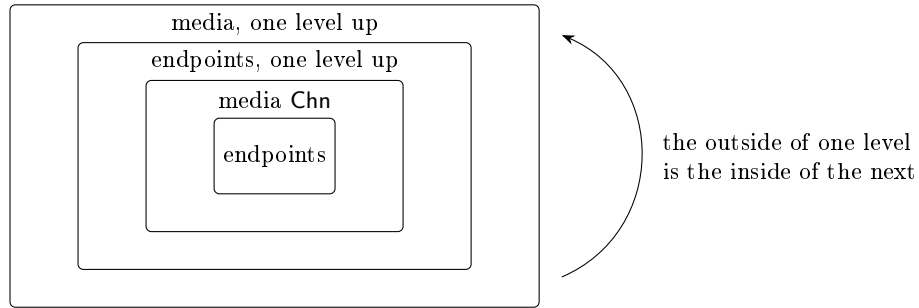


Figure 4: The generating step. Well-founded downward by the no-self-code theorem; bounded upward by the profitable radius.

Individuation is error correction. A region may close its namespace — become an individual — exactly when its boundary is rich enough relative to its radius: $h(B) \geq (\epsilon\theta/\rho)r$, with ϵ the per-hop error rate, ρ the repair throughput and θ the toll. Polynomial growth recovers a critical radius $r^* = \sqrt{\rho/\epsilon\theta}$; an expander makes it linear. Nature rides the boundary between centralisation and decentralisation because centralisation compresses and decentralises poorly under error, and error correction is the principle that says when to close over a namespace.

Individuals are porous, and porosity is not optional. Perfect closure is death: no harvest, no assay. An individual has a closed interior namespace and an open boundary namespace, and the boundary channels must have redex pathways to the interior or they are stranding sites. Deep integration is exponentially fragile in the number of hops, which is why ownership is what buys depth. Individuation is a fixed point, and the circularity is the fractality.

What this costs in copies. Following §35: a multi-skilled individual is a composite of ecologies-as-learners, each skill an ecology in its own right, and the copy number is a series indexed by stratum rather than an integer. Depth is exponentially fragile and width is linear, so buying depth obliges you to buy copies — the two axes are not orthogonal, which corrects an earlier statement in this corpus.

Observation 41.1 (The loop closes). *The individual at the top of one tower is the scientist at the bottom of the next. There is no distinguished level, and no level at which the observation discipline stops mattering: each stratum has its own namespace, its own budget, and its own logic generated from its own term structure.*

42 An aggregate mind assembles several disciplines

In plain terms: a mind does not have one logic. It has several, running side by side, seeing different things — and this is not an accident of biology but something the framework forces.

Everything in §39 composed learners that shared an observation discipline and differed only in what they knew. Nothing requires that. A learner is a distribution over a population, and different sub-populations may carry different dial settings. Parallel composition does not care.

42.1 Two games

Consider the card game *Set*. Each card carries four attributes with three values each, and a legal set is three cards on which every attribute is either all-the-same or all-different. What does it take to see one? It takes counting occurrences of abstract categories: three reds, three distinct shapes, three distinct textures. And counting is exactly what a fully structural discipline cannot do.

Observation 42.1 (Contraction is what makes counting impossible). *Contraction corresponds to idempotence of the container (Principle 7.1). A learner whose witnesses are collected in a set silently discards multiplicity: it can see that there are red ones, and it cannot see that there are three. Seeing three requires a container that withholds contraction — a bag — and the logic that container generates is the multiplicative fragment of §9. Set is a game one can only play in the linear discipline.*

Now consider poker. A player needs the same counting — outs, pot odds, how many of a rank remain — and needs something else entirely: to read the table. Who is playing with whom, which subset of players could between them take the pot without the rest, whose betting is advocacy and whose is noise. That is not a question about multiplicity. It is a question about which coalitions are actionable, and the discipline that generates it is the semitopological one of §10.

A poker player is therefore a learner that composes two observation disciplines with *different containers*. And here the framework says something sharper than “that is convenient”.

Observation 42.2 (The disciplines cannot be merged, only composed). *There is no single container that is both. Merging would require a distributive law between the two collection monads, and the no-go theorems for distributive laws turn on exactly the equations that separate them — commutativity and idempotence [38], the same two coordinates Principle 7.1 is built on. So an aggregate mind cannot have one unified logic that sees everything its parts see. It must run several disciplines in parallel and let them exchange results through a shared namespace, which is composition in the sense of §39 and is available with no new syntax.*

We would rather understate this than overstate it, so: the no-go theorems are about specific pairs of monads and their proofs are not uniform, and we have not checked the particular pair at issue. What is safe to say is that merging containers is exactly the operation those theorems obstruct, that the obstruction is generic, and that the framework therefore predicts modularity rather than having to assume it. If that is right, then the differentiated capabilities one observes in minds — a faculty for quantity, a faculty for social reading, a faculty for space — are not an engineering compromise inherited from evolution. They are what an aggregate observer has to look like.

42.2 Capabilities are separately present

Disciplines are separately generated, separately priced, and separately acquired. It follows immediately that they can be separately *absent*, and that the absence has a characteristic signature.

A learner missing a discipline is not globally impaired. It is precisely blind to the distinctions that discipline generates, and fully competent on the rest. By Principle 35.1 it also does not experience the blindness as blindness: it reports a world with fewer kinds in it than the world has, and the report is internally consistent. And by the legibility results of §34, what such a learner loses in a population it cannot read is *coverage*, not accuracy — it declines to answer rather than answering wrongly, which from the outside looks like reticence and from the inside looks like a world in which those questions simply do not arise.

Remark 42.3 (A speculation, flagged as such). It is tempting to read some of the observed variation in human social cognition this way — to conjecture that what differs is the availability or the weighting of something like the coalition discipline, rather than any general capacity, and that this would explain why such variation so often coexists with unimpaired or superior performance on exactly the counting-and-structure disciplines the same framework generates.

Two things should be said immediately. First, the framework admits a second reading of the same phenomenon, and it is not the deficit reading: two learners may both have a coalition discipline and disagree about which coalitions count as actionable — a different container, not a missing one. That reading predicts *symmetric* mutual illegibility rather than a one-way deficit, since each population is reading the other with the wrong opens, and §34’s counter-move result says the failure would show up as loss of coverage on both sides rather than error on either. The empirical literature contains a well-known argument of exactly that shape, and we note that the framework’s natural prediction sits with it rather than against it.

Second, we are in no position to adjudicate. This is a conjecture stated in the framework’s vocabulary to show that the vocabulary has empirical bite, and it is not a clinical claim; nothing in this document was built to make claims about people. We record it because a reader will think of it, and because the two readings are genuinely distinguishable — one predicts asymmetric failure, the other symmetric — which makes it a question that could be settled rather than argued.

42.3 The composite is a learner again

Note finally that this is the fractal of §41 in another costume. A poker player composing a counting discipline with a social one is a learner whose sub-learners run different logics, exchanging through the problem namespace — which is where, by Proposition 39.2, the positive-sum lives. The composite is itself a learner, with its own budget and its own confinement, and it can be composed again.

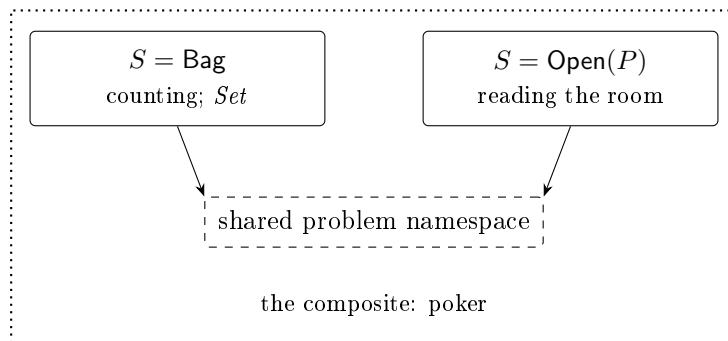


Figure 5: Two disciplines, two containers, one learner. There is no container that is both (Observation 42.2), so the composition happens at the level of learners.

43 What a learner can afford to see

The document ends where it must, which is in epistemology rather than engineering.

Every result above is a result about a *mortal* observer: one with a purse, whose observations are priced by depth, whose refutations are budget-relative, and whose undetermined verdicts have two sources it cannot tell apart. Such an observer does not have a logic in the sense of a fixed vocabulary handed down. It has an observation discipline, which is generated from the machinery it is made of and paid for out of the same budget as its food.

The consequences compound in one direction.

Its senses are the structural predicates it can afford, so it perceives shapes to a certain depth and no further. Its lookahead is the modal nesting it can afford, which by §31 is usually a poor purchase. Its hypotheses are formulae in the relaxation lattice, so its theories are confined to the ball it began in unless recombination or composition move it. Its counting is graded by its resolution, so it reports fewer kinds than the world contains — and, by Principle 35.1, does not see that it is doing so. Its refutations run out when its purse does. And the very distinctions available to it are fixed by dials it did not choose and cannot inspect from inside.

That is Kant’s picture with a ledger attached. The categories are not transcendent and they are not arbitrary: they are what a mortal, interacting learner can afford, generated from what it is made of. The framework’s contribution is not the observation — philosophers have had it for two centuries — but the arithmetic: the budget constraint can be *measured*, and the difference between what a learner sees and what is there can be computed.

Remark 43.1 (On the effectiveness of reason). There is a well-known puzzle about why mathematics should be so effective on the natural world, and on this account the puzzle is asked across a gap that is not there. The hypothesis language is generated from the substrate, and adequacy

— where it holds — makes it separate exactly what interaction separates. So a formal language is effective on exactly the domain to which interaction reaches, because it is the shadow that the access relation casts. Two consequences follow that are not merely restatements: transfer to non-ancestral domains should be expected, because adequacy is for interaction *as such* rather than for any particular interaction; and there should be a predicted boundary, at any relation that is not interaction across a cut. Both are stated in [7] and neither is established here.

Part VIII — Shown, not shown, open

An omnibus is easy to over-read. This part separates what has been established from what has been assembled, from what has merely been proposed, and lists what is open.

44 What is shown

1. **Two of the three components of a logic over a term language are determined by its presentation.** The structural connectives are the constructors lifted; the behavioural modalities are the context-labelled transitions lifted. Neither involves a design decision once the presentation is fixed (§2). This is the content of the claim that the logics here are not *ad hoc*, and it is the one claim in the document that is fully established rather than assembled.
2. **The collection container carries the substructural hierarchy.** Commutativity of the container’s aggregation is exchange; idempotence is contraction; the unit laws are weakening (Principle 7.1). A finer axis — which lattice operations the container makes primitive — separates intuitionistic, coalitional and linear settings that the coarse axis cannot (Principle 8.1).
3. **Classical linear logic is a setting of that dial.** Phase semantics is a two-layer container: an inner commutative monoid supplying the multiplicative, an outer lattice supplying the additive, and a pole supplying involutive negation (Observation 9.1). Spatial logic’s separating conjunction and linear logic’s $\otimes$ are the same operation (Observation 9.3).
4. **Coalition logic’s distinctive structure is predicted from container closure alone.** A meet-free container forces a relational residue in place of conjunction, and that residue is the compatibility relation of Gabbay’s semiframes (Observation 10.1), arrived at independently.
5. **The observation reading of a generated logic requires no proof theory** (Proposition 12.1), which is why the range of logics practically available for observation exceeds the range available for synthesis.
6. **Weight maps are graded clauses in disjunctive normal form** (Theorem 16.1), so the partition discipline they required is unnecessary and state-dependence is free (Proposition 16.3).
7. **Which-path erasure is decided by the continuation, not the weights** (Theorem 17.2, Corollary 17.3), and needs no new term former.
8. **Weighing is not selecting**, the two are the quantifier and selection monads, and attainment of a section is a condition on the value algebra (Proposition 21.1) — which independently picks out the algebra the coalition row needs (Observation 10.4).
9. **Choice strength can be written as clause shape**, one schema per Weihrauch degree in a single syntactic slot (Table 5); the extractable fragment and the choice-free fragment coincide (Corollary 23.2); and choice type is not preserved by reduction unless the **where** is stratified (Propositions 27.1, 27.3).
10. **The measured results of Table 8**, each produced by a simulation shipped with its note.

45 What is assembled rather than established

Several of the more striking statements in this document are syntheses across notes rather than theorems, and it should be possible for a reader to tell which.

- **The Girard-quantale criterion for negation in the graded sort** (Observation 9.7). A short step from facts available since [18], but the identification of the graded design's two-sortedness with the failure of V to be a Girard quantale is made here for the first time and has not been checked against every value algebra in use.
- **The five-witness account of negation** (§19). The five items are individually established in their sources; the claim that they are one phenomenon is an organising thesis.
- **The fibration reading of the hypothesis space** (Observation 37.1). The base and the fibre are each established; that their combination is the right way to describe a graded learner is a proposal, and the coherence conditions on the fibration have not been checked.
- **The identification of persistence with the exponential** (Remark 9.6). Deliberately left as a conjecture.
- **The modularity argument** (Observation 42.2). That merging two containers is the operation the no-go theorems obstruct is right; that the obstruction holds for the particular pair at issue has not been checked, and the theorems are not uniform.

46 What is not shown

- **Adequacy is not general.** A generated logic separates exactly what interaction separates only when the target specification supplies enough structure. Many useful settings do not (§3).
- **Decidability is not general.** Model-checking the ideal logic does not terminate; the practical discipline is to restrict the fragment or the connectives, at the cost of adequacy, and to bound the rest with a budget.
- **The biology is not established.** The swarm results are untested predictions; the framework does not out-predict evidence-accumulator models on trajectories and is not a competing dynamical model.
- **The ecological results are model results.** They are exact over their stated worlds, and their pricing schedule is a choice that was varied. Band shapes are robust to it; boundaries are not.
- **The quantum reading is confined.** Unmodified ρ interferes only on indistinguishable racers and only constructively; the graded language reaches **PP** unless clauses are constrained to be isometric; and no claim of quantum advantage is made or tested.
- **Binding is scoped out** of the constructor-extension construction that reconciles spatial strictness with adequacy, which remains its largest gap.

47 Negative results, kept

Deliberately retained, because a summary that discards them misrepresents the programme.

- **Grading can buy nothing.** At coalition size $k = 13$ the tropical min is zeroed by a single dissenter, so crisp and graded rungs return identical rows. Grading pays only if the size threshold is relaxed too: two dials, not one.
- **Cross-inhibition has an unexpected cost.** It suppresses correct advocacy along with incorrect, so the swarm picks the better site slightly less often with it than without (.7235 against .7845). Not predicted; open.
- **The correct theory is a bad buy.** χ is dominated on the noughts ladder, optimal in two cells of the phase grid, and its invasion threshold is an endowment above 12,474 price units.
- **Depth can make a learner worse.** Deepening the theory while holding the opening fixed moves win rate from .833 to .548.
- **Confidence does not raise the mean.** It reduces variance. The crisp forager has the highest terminal wealth conditional on survival, and the lowest survival.

48 Open problems

01. **The two gradings.** Prove or refute Conjecture 26.1. The obligation is to make “the candidate container” precise for clauses that use the graded modality, where candidates at one site depend on transitions available at another.
02. **The exponential.** Settle Remark 9.6: is persistence the calculus’s native!? The statement to prove concerns contraction on received data in the presence of reflection.
03. **Isometry as a static judgement.** Decide isometry for clauses using the graded modality, or show it undecidable. Until this exists, a graded program in coherent mode should be read as post-selected.
04. **The converse of the Diaconescu cost.** We have that a choice-realising resolver exhibits no interference deficit. The converse would need a topos of graded predicates.
05. **Length-asymmetric reconvergence.** Does reflection supply a second, non-local source of interference, by making the residual step count depend on which racer won?
06. **Testimony with two coordinates.** Carry strength and confidence as a pair with a lax algebra and a canonical aggregation order, and settle the plagiarism equilibrium as a condition on the grading vector (Observation 40.1).
07. **Coherence of the fibration.** State and check the coherence conditions for the hypothesis fibration of §37, and give a search procedure that moves in both coordinates with a principled exchange rate between a fibre step and a base step.
08. **Which open the bees act on.** The discriminating experiment is whether switchers — scouts in two advocacy sets — are over-represented among bees that neither pipe nor stop-signal. It tests the logic rather than the arithmetic.

- O9. **Learning the semitopology.** An observer must discover the set of actionable coalitions rather than being given it. Identifying a topen is quadratic, and a topen-boundary error is a shape error, unreachable by refinement.
- O10. **Heterogeneous quorums** and the compatibility relation as a cheaper observation than the modality itself.
- O11. **The non-commutative row.** What does the generator emit when the container is `List` — for a term language with genuine sequence structure — and does it recover the Lambek calculus in the same sense that §9 recovers linear logic?
- O12. **Binding.** Extend the constructor-extension construction to binders, which is the acknowledged gap in the reconciliation of spatial strictness with adequacy.
- O13. **Modularity as a no-go theorem.** Settle Observation 42.2 for the specific containers at issue — is there a distributive law between the bag monad and the opens of a semitopology? A negative answer would turn the modularity of an aggregate observer from a prediction into a theorem.
- O14. **Exchange between disciplines.** An aggregate mind composing several disciplines must translate between them across the shared namespace. What is the type of that translation, what does it cost, and what is lost — given that the containers, by hypothesis, do not merge?

49 Coda

The thesis of this document can be put in one sentence, and it is worth putting badly first so that the good version is legible.

The bad version: *logic is relative to the machine*. That sounds either trivial or nihilistic, and it is neither.

The good version: *a logic is an observation discipline, generated from a term language by four parameters, and a mortal thing does not choose its discipline freely — it can afford some settings and not others, and the settings it can afford determine what distinctions exist for it*. The four parameters are the signature, the rewrite rules, the container in which witnesses are collected, and the algebra in which observations take their values. The first two are given by what the thing is made of. The third says what kind of world it takes itself to be in — a world of sets, of resources, or of coalitions that can act. The fourth says what an observation returns, and it is the one this document argues is undersupplied in practice: almost everything is built with $V = \mathbb{B}$, and a learner built that way has, as we have seen measured, no coordinate in which to record disappointment.

The instrument is small. It is a formula in a binder, with a value. What it does is put the resolution of a system’s own non-determinism inside the language, where it can be reasoned about, priced, revised, and — the point of the whole exercise — *learned*.

And one consequence is worth leaving the reader with, because it is where the account stops being about programming languages. If the containers do not merge, then no observer of any size has a single logic. An aggregate mind is an assembly of observation disciplines: one that counts, one that reads a room, one that reads space, each generated from the same term structure by a different setting of the dials, each separately priced, each separately acquirable — and

therefore each separately absent. That is a claim about architecture, and it predicts exactly the differentiated and dissociable capabilities that minds are in fact observed to have. A mind is not one instrument. It is a workshop, and the interesting question about any given one is which instruments are on the bench.

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result	figure	source
Non-modal rungs dominate the ladder	92.3% of yield for 34.3% of cost	[8]
The correct theory is dominated	.873 at 679.2 vs .918 at 255.9	[8]
Confinement, measured	best opening reverses; .833 $\rightarrow$.548	[8]
Affordability inverts with the world	Nim returns rising: +.00602/ + .01262/ + .01476	[9]
Bootstrap trap	0.00 openings, 520.0 vs 3543.7	[2]
Crisp guard cannot learn to stop	survival .0029 in the harsh world	[2]
Confidence buys variance, not yield	.8501/4084.7 vs .9485/3543.7	[2]
Scepticism has an interior optimum	$k^* = 9$; ensemble $k^* = 64$	[2]
Interleaving inflation is real	exactly 2, 6, 24 at $n = 2, 3, 4$	[2]
Shor as deliberate interference deficit	12 of 16 outcomes at amplitude 0	[2]
The modality buys irreversibility	precision 1.0000 at 0.37% coverage	[5]
Illegibility denies liveness, not safety	coverage .7520 $\rightarrow$.0115, precision held	[5]
The margin is a clock	42.4/29.6/28.9/18.0 ticks by quartile	[5]
The $2k$ theorem	split 0.0000 below $S = 30$; 0.0130 at 30	[5]
Cross-inhibition changes jobs	deadlock .5120 $\rightarrow$.0000; split .3860 $\rightarrow$.1170	[5]
A coarse observer under-reports kinds	1 kind of 4; count inflated $3.4\times$	[11]

Table 1: The headline measurements. Worked in Part VI.

V	$\oplus$	$\otimes$	reading
$\mathbb{B}$	$\vee$	$\wedge$	crisp: the classical guard
$[0, 1]$	$\max$	$\times$	possibility; probabilistic logic
$\mathbb{R}_{\geq 0}$	$+$	$\times$	propensity; stochastic simulation
Viterbi ($[0, 1]$)	$\max$	$\times$	best explanation, with a witness
tropical ($\mathbb{R} \cup \{\infty\}$)	$\min$	$+$	least cost, with a witness
$\mathbb{C}$	$+$	$\times$	amplitude; interference
a quantale	$\bigvee$	$\otimes$	graded truth with fixed points

Table 2: Value algebras used in this document. The last row is developed in Part II, where it turns out to be the same object as the container of classical linear logic.

logic	Σ, R	S	V	where it appears
namespace / spatial-behavioural logic	rho calculus	P_{fin}	$\mathbb{B}$	senses and assays [7]; the classifier notes
intuitionistic (geometric)	any	frame of opens	$\mathbb{B}$	the ambient reading of affirmation [32]
classical linear logic	any	$P_{\text{fin}}(\text{Bag } A)$, with a pole	$\mathbb{B}$	§9; resource discipline [17]
Gabbay’s coalition logic	participants	$\text{Open}(P)$, a semiframe	$\mathbb{B}$	consensus [33, 6]
the graded coalition modality	rho calculus	$\text{Open}(P)$	tropical / Viterbi	herd and swarm [5]
graded where-clauses	rho calculus	P_{fin}	$[0, 1]$, $\mathbb{R}_{\geq 0}$, $\mathbb{C}$	the instrument [2]
quantale-graded rho	rho calculus	P_{fin}	a quantale	fixed points survive [14]
the choice-hierarchy formulae	rho calculus	Bag of candidates	any of the above	altitude and price [3]
weighted rewriting	any MeTTaIL GSLT	P_{fin}	$\mathbb{R}_{\geq 0}$, $\mathbb{C}$	simulation [12]

Table 3: Logics as settings of the four parameters.

placement	judgement	needs	who uses it
specification	$\vdash \varphi \triangleright P$	a proof system	the designer
test	$P \models \varphi$	the semantics, plus a budget	the observer
guard	φ on the binder	per-match evaluation	the system itself
hypothesis	$\varphi_t \rightsquigarrow \varphi_{t+1}$	a lattice of formulae	the learner

Table 4: Four uses of one formula.

degree	clause shape	remark
$\perp$	fixed-point-free	no choice demanded
$\text{LLPO} \equiv \text{C}_2$	two-message race, Σ_1^0 guards	rho programmers write this by accident
$\text{C}_{\mathbb{N}}$	$\mu X.([\beta] \oplus \langle K \rangle X)$	the width engine
WKL	$\nu X.([\beta] \otimes \langle K \rangle X)$	the depth engine; compactness
WWKL	the $\mathbb{R}_{\geq 0}$ row	Las Vegas; the one point where the two axes touch
lim	plasticity acting on the limit	learning is free; knowing you have finished is the jump
DC	plasticity itself	no new syntax; an ω -fold product
$\text{C}_{\mathbb{N}^{\mathbb{N}}}$	both engines	

Table 5: Clause shapes by Weihrauch degree.

rung	win rate	cost	return per unit
0 (random)	.437	0	—
1 (win now, non-modal)	.668	40.9	+.0057
2 (block, non-modal)	.798	77.8	+.0035
3 ($\exists s. \langle \langle K_s \rangle \rangle \text{Fork}$)	.831	134.9	+.0006
4 ($\exists s \forall t$, alternating)	.836	245.7	+.00005
5 (spatial, name-level depth 0)	.918	255.9	+.0082
χ (exact)	.873	679.2	−.0001

Table 6: The noughts-and-crosses ladder [8]. Rungs are ordered by formula complexity, not by performance.

observer resolution	kinds reported	count inflation
0	1	$3.4\times$
2	—	$2.6\times$
3	all	$1.0\times$

Table 7: What a coarse observer reports [11]. The copy series is (1152, 48, 12, 1); its flat collapse is 1213.

result	figure	source
Non-modal rungs dominate the ladder	92.3% of yield for 34.3% of cost	[8]
The correct theory is dominated	.873 at 679.2 vs .918 at 255.9	[8]
Confinement, measured	best opening reverses; .833 $\rightarrow$.548	[8]
Affordability inverts with the world	Nim returns rising: +.00602/ +.01262/ +.01476	[9]
Bootstrap trap	0.00 openings, 520.0 vs 3543.7	[2]
Crisp guard cannot learn to stop	survival .0029 in the harsh world	[2]
Confidence buys variance, not yield	.8501/4084.7 vs .9485/3543.7	[2]
Scepticism has an interior optimum	$k^* = 9$; ensemble $k^* = 64$	[2]
Interleaving inflation is real	exactly 2, 6, 24 at $n = 2, 3, 4$	[2]
Shor as deliberate interference deficit	12 of 16 outcomes at amplitude 0	[2]
The modality buys irreversibility	precision 1.0000 at 0.37% coverage	[5]
Illegibility denies liveness, not safety	coverage .7520 $\rightarrow$.0115, precision held	[5]
The margin is a clock	42.4/29.6/28.9/18.0 ticks by quartile	[5]
The $2k$ theorem	split 0.0000 below $S = 30$; 0.0130 at 30	[5]
Cross-inhibition changes jobs	deadlock .5120 $\rightarrow$.0000; split .3860 $\rightarrow$.1170	[5]
A coarse observer under-reports kinds	1 kind of 4; count inflated $3.4\times$	[11]

Table 8: The headline measurements, repeated from Table 1 for readers who came straight to this part. Every figure is produced by a simulation shipped with its note.